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PLATING POSTERS INC • METAL FINISHING REFERENCE SERIES

INTEGRAL COLOR ANODIZING ONE-STEP COLOR

PROCESS VARIANT **INTEGRAL COLOR**

= COLOR MADE BY THE ANODIZE ITSELF

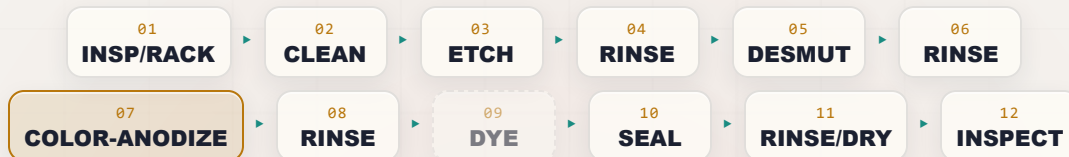
NO DYE

ARCHITECTURAL • COLOR FROM THE ANODIZE ITSELF

SULFURIC + ORGANIC ACID • BRONZES / GRAY / BLACK • LIGHTFAST • AAMA 611 CLASS I

THE COMPLETE TRAINING MANUAL

*A full training course for the brand-new operator — from “wait, the part is the positive electrode?” to a signed certificate of completion. This is the architectural anodize where **the color is grown right into the coating in one step** — there is **no dye**. As the oxide grows in a special sulfuric-plus-organic-acid bath, the alloy and the process produce **inherent bronzes, grays, and blacks** that are locked into the ceramic and will not fade for decades. The trade-off: it takes **more voltage, more cooling, more amp-hours, and a tightly controlled alloy** than ordinary dyed work. Assumes you know nothing. Teaches you everything.*



OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 • 2026 • BEGINNER-FRIENDLY STYLE

Training reference only. Always verify exact parameters against your chemistry supplier's Technical Data Sheets (TDS), customer/architectural specifications (e.g., AAMA 611, ASTM B580 architectural Type A, ASTM B137/B244, and applicable customer/project specs), and current Safety Data Sheets (SDS), and comply with OSHA, EPA, REACH, and local regulations before production use. **Integral color uses sulfuric acid plus organic acids, runs at higher DC voltage with greater heat load, and shares the caustic-etch, hydrogen, and near-boiling-seal hazards of every anodize line.**

— FIND YOUR WAY

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REMEMBER

This manual is a training and reference tool — **not** a substitute for your facility’s written procedures, your supplier’s TDS/SDS, your customer’s spec (AAMA 611, ASTM B580), or your supervisor’s instructions. When this manual and your shop’s procedure disagree, **follow your shop’s official procedure and ask your supervisor.**

WELCOME ABOARD

IF YOU DON'T KNOW AN ANODE FROM AN APRON, YOU'RE IN THE RIGHT PLACE — HERE THE WHOLE TRICK IS MAKING *COLOR OUT OF THE ANODIZING ITSELF*, WITH NO DYE

Welcome. If you are holding this manual, you are about to learn how to run — or at least understand — an **integral color anodizing line**, the process the trade also calls **one-step color**, **self-color**, or by the old brand names **Kalcolor** and **Duranodic**. Maybe you started this week; maybe you've racked parts for a year and finally want to know *why*. Either way, you're in the right place.

Here is the first big thing to wrap your head around, and it'll feel backwards if you've seen electroplating: **in anodizing, we are not coating your part with a different metal. We are growing a tough, ceramic oxide layer out of the aluminum itself.** And to do that, your part is the **anode** — the *positive* electrode — the exact opposite of plating. That's the single biggest *process* idea in this whole book.

Here is the second big idea, and it is the reason this process exists at all: **on an integral color line, the color is made BY the anodizing — there is no dye step.** On an ordinary decorative line, you anodize a clear coating and then soak colored dye into its pores. Here, we run a **special electrolyte (sulfuric acid plus an organic acid)** at **higher voltage** and grow a **colored oxide directly** — champagne, bronzes from light to dark, gray, and black — with the color built right into the ceramic. Because there is no dye to fade, the color is **extraordinarily lightfast and abrasion-resistant**. That is exactly why this is **the** anodize on building facades, curtain walls, storefronts, and window frames that have to look right after thirty years of sun and weather.



REMEMBER

Do not read “no dye, super durable” as “easy.” Integral color buys its durability with **more energy, more cooling, longer cycles, and a much tighter grip on the alloy.** The color is set by **coating thickness (amp-hours)** and by the **exact alloy and temper** of the part. The line still has **strong acid, hot caustic etch, flammable hydrogen, high-current DC, and a near-boiling seal.**

The third thing up front: **this manual assumes you know nothing about anodizing, chemistry, or electricity.** That's not an insult — it's a promise. Every term gets defined the first time we use it; every step explains not just *what* but *why*. Don't memorize it; *understand* it.

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, station by station, in the order a part actually travels: inspect/rack, clean, etch, rinse, desmut, rinse, **color-anodize** (sulfuric + organic acid, part = anode, high voltage), rinse, **(no dye)**, seal, rinse/dry, inspect. Reading it in order matters: **the order of an anodizing line is not random** — and neither is this book's.



TIP

Read the Fundamentals (Part One) through *before* any station chapter — **especially Chapter 3 (the three coloring routes) and Chapter 7 (alloy & temper).** The station chapters assume you already grasp the big picture: the part is the anode, oxide grows out of it, **the color comes from the coating itself**, and the color depends on **how thick you grow it and what alloy you grow it on.**

THE ICONS — YOUR FRIENDLY MARGIN HELPERS



TIP

A shortcut or trick of the trade — the kind of thing a 20-year veteran would lean over and tell you.



REMEMBER

A core idea worth locking in — the load-bearing ideas. If you forget everything else on the page, don't forget these.



HEADS-UP

A safety warning. Read every single one — these protect your skin, eyes, lungs, and long-term health. We use them liberally.



TECHNICAL STUFF

Deeper chemistry for the curious. Optional — skip it and you can still run the line.



FUN FACT

A bit of trivia to make the science stick (and to give you something to say at lunch).

TWO MORE THINGS YOU'LL SEE AT EVERY STATION

Each station chapter ends with a **Teach-Back** checklist (often paired with a red “Top Mistakes” card) and a bordered **Sign-Off** box your *trainer* initials once you've shown the skill safely. The Teach-Back is your study guide; the Sign-Off is the gate.

WHAT YOU'LL BE ABLE TO DO

LEARNING OBJECTIVES

MASTER THIS AND EVERY STATION CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

1. **Explain what anodizing is** in plain language — and how it is fundamentally *different from electroplating*: the part is the **anode**, and we **grow an aluminum-oxide layer out of the aluminum itself** instead of depositing a different metal onto it.
2. **State what makes integral color different**: the color is produced **by the anodizing itself, in one step, with no dye**. A sulfuric-plus-organic-acid electrolyte run at higher voltage grows a **colored oxide** (champagne, bronzes, gray, black) whose color is **locked into the ceramic — exceptionally lightfast and abrasion-resistant**.
3. **Explain how the color forms** — finely dispersed alloying constituents and the organic-acid-modified hard oxide **scatter and absorb light**, and **color deepens as the coating gets thicker**. Color is therefore controlled by **amp-hours / thickness**, and is repeatable only when **alloy + temper + parameters** are tightly controlled.
4. **Explain why integral color is the architectural anodize** — facades, curtain walls, storefronts, window/door frames — because the color survives decades of UV and weather, and recognize where it sits under **AAMA 611** and **ASTM B580 / architectural Class I ($\geq 18 \mu\text{m}$ / 0.7 mil)**.
5. **Compare the three coloring routes** — **dyeing** (cheap, any color, least lightfast), **integral color** (durable/lightfast, limited bronze-to-black palette, energy-intensive), and **two-step electrolytic** (durable, consistent, the modern architectural standard) — and say when each is chosen.
6. **Explain dimensional growth** — the coating grows roughly half in / half out; because integral color coatings are *thick* (Class I), the growth is real and must be planned for on tight-tolerance work.
7. **Place integral color in the family** — Type I (chromic), Type II (sulfuric, dyeable), Type III (hardcoat), and the specialty routes (BSAA, PAA, bright, integral color, two-step electrolytic).
8. **Explain why alloy and temper matter more here than anywhere else** — the alloying constituents *are* the colorant, so different alloys/heats/extrusions color differently, making **lot-to-lot and sheet-vs-extrusion color matching** the central challenge of the job.
9. **Name the stations of the line in order** and explain *why* the order can't be rearranged (and why there's no dye step).
10. **Recognize the main hazards** — sulfuric + organic acids, **higher-voltage DC (greater shock/arc energy)**, **more process heat / cooling load**, hot caustic etch, flammable hydrogen, and a near-boiling seal — and know the correct PPE, emergency response, and waste handling.

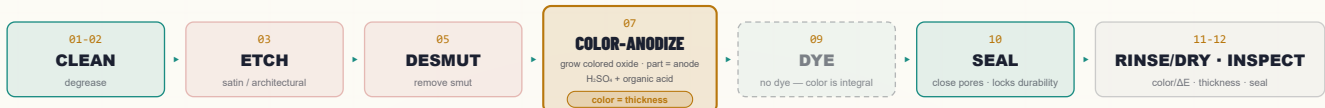


REMEMBER

You are not expected to hit all of these on day one. Anodizing is a craft. But the *safety* objectives (#10) are not optional and not gradual — you must have those before you work the line at all.

• THE LINE AT A GLANCE

Don't memorize yet — feel the rhythm: **Inspect/Rack** → **Clean** → **Etch (satin)** → **Rinse** → **Desmut** → **Rinse** → **COLOR-ANODIZE (sulfuric + organic acid, high V)** → **Rinse** → **(no Dye)** → **Seal** → **Rinse/Dry** → **Inspect**.



REMEMBER

Notice the pattern: **a rinse follows almost every working tank**. Rinses aren't breaks — they're checkpoints that keep one chemistry from poisoning the next. And notice the one big difference from a decorative line: **there is no dye tank**. The color is born in the anodize tank itself.



FUN FACT

The bronze and black aluminum you see on classic mid-century and modern office towers, bank buildings, and storefront systems is, in huge part, **integral color anodize**. When architects in the 1960s–80s wanted aluminum that would hold a rich bronze for the life of the building, this is the finish they specified — and much of it still looks the way it did the day it went up.

CHAPTER 1

WHAT IS ANODIZING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT ANODIZING IS

THE ONE-SENTENCE VERSION

Anodizing is using electricity to grow a hard, protective oxide layer out of the surface of an aluminum part — the part itself becomes part of the coating. That's the whole idea. Now let's unpack it, because it's genuinely different from plating and that difference trips up almost everyone at first.

FIRST, WHAT PLATING DOES (SO YOU CAN SEE THE DIFFERENCE)

In electroplating — zinc, nickel, chrome — you take your part, hook it up as the **cathode** (the *negative* electrode), and you *deposit a different metal* onto it from a bath. The coating is a *separate metal sitting on top of* your part.

Anodizing flips almost all of that:

- Your part is the **anode** — the **positive** electrode. (That's literally where the word “anodize” comes from.)
- We don't add a different metal. We take the **aluminum that's already there** and convert its surface into **aluminum oxide** by feeding electricity through an acid bath.
- The coating isn't sitting *on* the part — it **is** the part's surface, chemically grown and locked in. It can't peel like paint or flake like a bad plate, because it's integral to the metal.



REMEMBER — THE SINGLE BIGGEST IDEA IN THIS BOOK

In plating, the part is the **cathode** and you **add** a metal. In anodizing, the part is the **anode** and you **grow oxide out of the part itself**. If you remember nothing else from Chapter 1, remember this. Everything else follows from it.

HOW IT PHYSICALLY WORKS: THE KITCHEN-SINK PICTURE

Picture a tank full of liquid called the **electrolyte** (or just “the bath”). For integral color, that liquid is **dilute sulfuric acid blended with one or more organic acids** (we'll meet them in Chapter 2). You hook two things to a power supply called a **rectifier** (it converts ordinary AC wall power into the steady one-direction “DC” current anodizing needs):

- **The anode** — this is your **aluminum part**, connected to the **positive** terminal. “*Anode = the part, the thing growing oxide.*”
- **The cathode** — bars or plates (commonly **lead** or **aluminum 6063**) hanging in the same tank, connected to the **negative** terminal. The cathode just completes the circuit and is where **hydrogen gas** bubbles off.

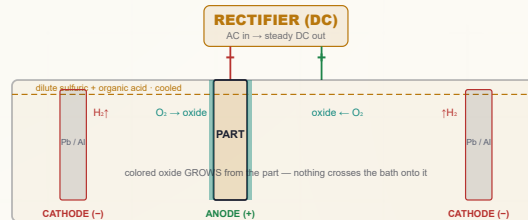


TECHNICAL STUFF — THE ORGANIC ACID'S JOB

On a plain Type II line, dilute sulfuric grows a clear, dyeable coating. Adding **organic acids** (sulfophthalic, sulfosalicylic, maleic, oxalic) changes how the oxide grows — producing a **harder, denser film that develops color as it grows**. Combined with the alloy's own constituents, that's where the bronze comes from. Exact identities and concentrations are **proprietary to each process — read your TDS**.

When you turn on the rectifier, here's the dance that happens:

1. Current flows. At the surface of your aluminum part (the anode), water in the bath is split and **oxygen** is delivered right at the metal surface.
2. That oxygen immediately combines with the surface aluminum to form **aluminum oxide (Al₂O₃)** — a hard, ceramic layer.
3. The acid is slightly aggressive, so as the oxide forms, the acid is *also* gently dissolving it — and this tug-of-war creates the famous **porous structure** (Chapter 4). On an integral color line the **organic acid changes how this tug-of-war runs**, producing a harder, denser, **colored** oxide.
4. Over at the cathode, **hydrogen gas** bubbles off into the air (a safety item — flammable gas — we'll come back to it).



Compare to a plating cell: the labels are reversed — here the **part is the ANODE (+)** and the counter-electrodes are the cathodes.



TECHNICAL STUFF — THE TWO REACTIONS

At the part (anode), aluminum is oxidized: simplified, $2\text{Al} + 3\text{H}_2\text{O} \rightarrow \text{Al}_2\text{O}_3 + 6\text{H}^+ + 6\text{e}^-$. At the cathode, those electrons reduce hydrogen ions: $6\text{H}^+ + 6\text{e}^- \rightarrow 3\text{H}_2\uparrow$. So **oxygen builds the coating at your part; hydrogen gas leaves at the counter-electrode**. No aluminum travels through the bath onto something else — it stays put and turns to oxide in place.



TECHNICAL STUFF — WHY THE BATH MUST BE COOLED, AND WHY MORE HERE

This reaction is **exothermic** (it makes heat), and a lot of electrical energy turns to heat right at the part surface. On an integral color line we push **higher voltage and more amp-hours** to build a thick colored film, so the heat load is **larger than on a plain Type II tank**. If the bath warms, the acid dissolves the oxide too aggressively and you get a soft, powdery, “burned,” **off-color** coating instead of a hard, evenly colored one. That's why an integral color tank has **strong refrigerated cooling and vigorous agitation**. On this line, *cold, steady, and well-stirred is color consistency*.



FUN FACT

Anodizing is one of the few finishing processes where the coating is *grown from* the part rather than *added to* it — which is exactly why an integral-color bronze can't “chip off” to bare aluminum the way painted bronze can. The oxide and the metal are continuous, and the color is buried inside the oxide.

CHAPTER 2

WHAT IS INTEGRAL COLOR?

THE ONE-STEP ARCHITECTURAL COLOR – DURABLE, LIGHTFAST, AND GROWN RIGHT INTO THE CERAMIC

Integral color anodizing (also called **one-step color**, **self-color**, or by the historic trade names **Kalcolor** and **Duranodic 300**) is a sulfuric-acid anodize **modified with organic acids** that produces **color in the coating during anodizing — no dye is ever used**. As the oxide grows, two things make it colored:

1. **Alloying constituents** in the aluminum (fine intermetallic particles of iron, silicon, manganese, chromium, etc.) become **occluded in the oxide** and **scatter/absorb light**.
2. The **organic-acid-modified oxide** is harder and denser and develops a structure that itself **scatters light toward the bronze/gray end**.

The result is a coating that comes out of the tank already colored — **champagne, light to dark bronze, gray, and black** — with the color **part of the ceramic, not a layer on top**. Because there is no organic dye to break down in sunlight, the color is **extremely lightfast** and the film is **more abrasion-resistant than ordinary Type II**.

★

REMEMBER

Integral color's defining feature is right in the name: the color is **integral** — born inside the coating. **Darker color = thicker coating**. You don't pick a color off a dye shelf; you *grow* it by running more amp-hours, on the right alloy. That single fact drives everything about how this line is operated.

• WHY SHOPS (AND ARCHITECTS) CHOOSE IT

ADVANTAGE	WHAT IT MEANS ON THE FLOOR / ON THE BUILDING
Outstanding lightfastness	Color is in the ceramic, not a dye — it won't fade in decades of UV. The reason it's specified for facades.
Abrasion / wear resistance	The organic-acid oxide is harder and denser than plain Type II — survives handling, cleaning, weather.
Color can't peel or flake	It's integral — grown from the metal, buried in the oxide.
Architectural durability	Meets architectural Class I (≥18 μm) for long exterior service under AAMA 611.
Rich bronzes & blacks	The classic warm architectural metals — the look that defined a generation of buildings.

WHERE YOU'LL SEE IT

Building **curtain walls and facades, storefronts and entrances, window and door frames, column covers, fascia, louvers, sunshades, and trim** — and high-end durable goods that need a colored finish that will *never* fade. If you've stood in front of a bronze or black aluminum office building, you've almost certainly seen integral color.

TIP

When a print or architectural spec calls for “**bronze anodize, AAMA 611, Class I**,” or names a color like “**dark bronze / statuary bronze / black**” with no mention of dye, it is very often asking for **integral color or two-step electrolytic color** — *not* dyed Type II. Dyed bronze exists but is not durable enough for exterior architecture. **Check the spec and ask** — chemistry, cost, and durability all change.

FUN FACT

“Kalcolor” was Kaiser Aluminum’s name and “Duranodic” was Alcoa’s — both 1960s integral-color processes using organic-acid-modified sulfuric baths (sulfophthalic and sulfosalicylic acids). They became so common in architecture that for years “Duranodic bronze” was used almost as a generic color name on building drawings, the way people say “Kleenex” for tissue.

THREE WAYS TO COLOR ANODIZED ALUMINUM

THE ONE CHAPTER THAT MAKES SENSE OF WHERE THIS PROCESS SITS — DYE VS. INTEGRAL VS. TWO-STEP

This is the chapter that explains *why integral color exists* and when it's the right tool. Anodized aluminum can be colored **three fundamentally different ways**, and they are not interchangeable.

ROUTE 1 — DYEING (TYPE II + ORGANIC DYE IN THE PORES)

Anodize a normal clear coating, then soak the **open pores** full of **organic (or inorganic) dye**, then seal.

- **Pros:** Cheap, fast, **any color** including brilliant reds, blues, golds. Low energy.
- **Cons:** **Least lightfast** — many organic dyes fade in UV/outdoors. Color sits in the pores, not in the oxide structure. Best for **indoor / decorative / consumer work**.

ROUTE 2 — INTEGRAL COLOR (COLOR FROM THE ANODIZING ITSELF) — *THIS MANUAL*

Anodize in a **sulfuric + organic acid bath** at **higher voltage**; the **alloy constituents + the modified oxide** make the color as the coating grows. No dye. Seal for durability.

- **Pros:** **Extremely lightfast and abrasion-resistant** — color is in the ceramic. Excellent for **exterior architecture**.
- **Cons:** **Limited palette** — champagne, bronzes, gray, black (no bright colors). **Energy-intensive** (high voltage, long cycle, big cooling load). **Very alloy-sensitive** — color depends on the metal, so **lot-to-lot matching is hard**.

ROUTE 3 — TWO-STEP ELECTROLYTIC COLOR ("ELECTROCOLORING")

Anodize a normal clear coating, then move to a **second electrolytic tank** containing **metal salts** (tin, nickel, or cobalt) and apply **AC current** to **deposit fine metal particles deep in the bottom of the pores**. Then seal.

- **Pros:** **Durable and lightfast** like integral color (the color is metal, not dye), **but far more consistent** and **much lower energy** — and **less alloy-sensitive**. Tin gives champagne→bronze→black.
- **Cons:** Two process steps and a second tank; palette still mostly bronzes/black/gray (some blues/burgundies possible).
- **This is the modern architectural standard** and is **steadily displacing integral color** — it's cheaper and more repeatable. *(It gets its own manual in this series — we build it next.)*



REMEMBER — THE THREE ROUTES IN ONE BREATH

Dye = color *in the pores* (cheap, any color, fades). **Integral** = color *in the oxide itself* (durable, bronzes/black, energy-hungry, alloy-fussy). **Two-step** = metal *deposited in the pores* (durable, consistent, the modern architectural default). Integral color is the durable old-school route; you'll still see it specified for its proven track record and certain colors, but know that two-step often wins on cost and consistency.

	DYE (TYPE II)	INTEGRAL COLOR	TWO-STEP ELECTROLYTIC
Where the color lives	Organic dye in the pores	In the oxide itself (alloy + structure)	Metal particles deep in the pores
Lightfastness	Low–medium (UV fades)	Excellent	Excellent
Palette	Any color	Champagne, bronzes, gray, black	Champagne → bronze → black (some others)
Energy / cost	Low	High (high V, long cycle, cooling)	Moderate
Color consistency	Good (dye-controlled)	Hard (alloy-driven)	Very good
Typical use	Indoor / consumer / decorative	Exterior architecture (legacy & spec)	Exterior architecture (modern std)

**TECHNICAL STUFF — WHY INTEGRAL COLOR IS SO ALLOY-SENSITIVE AND TWO-STEP ISN'T**

In integral color, **the alloy's own intermetallic particles are part of the colorant** — so a different heat of metal, a different extrusion, or a different temper literally changes the color at the same amp-hours. In two-step, the color comes from **metal you deposit yourself** in a separate tank, so it's far less dependent on the base alloy. That's the core reason the industry drifted from integral toward two-step: **you control the color directly instead of relying on the metal to make it for you.**

**FUN FACT**

All three routes can produce a “bronze” that looks similar on day one. Put them in the desert sun for ten years and they sort themselves out: the **dyed** one fades first, while the **integral** and **two-step** bronzes hold — because in those two the color is durable metal/ceramic, not an organic dye.

**TIP — PICK THE ROUTE TO THE REQUIREMENT**

If a customer needs a **bright color indoors**, that's **dyed Type II**. If they need a **durable bronze/black outdoors**, that's **integral color or two-step**. When cost and lot-to-lot consistency matter most and the color is a bronze/black, **two-step** usually wins. Knowing which route fits the job is half of running this line well.

CHAPTER 4

COLOR FROM THE COATING

THE STRANGEST, MOST IMPORTANT IDEA ON THIS LINE — YOU GROW THE COLOR, YOU DON'T APPLY IT

This is the chapter that makes integral color *integral color*. Slow down here.

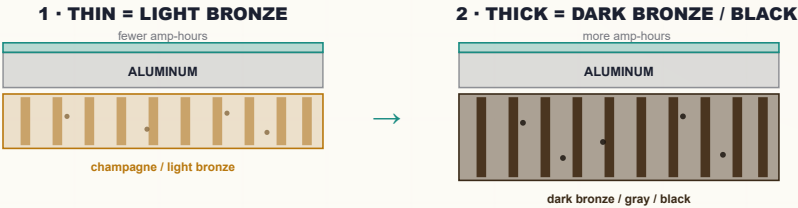
THE OXIDE IS STILL POROUS — BUT THE COLOR IS IN THE OXIDE, NOT THE PORES

Like all sulfuric anodizing, integral color grows a **porous oxide**: billions of tiny columns with a microscopic pore down the center of each, sitting on a thin dense **barrier layer** against the metal. But here's the twist — **we don't put color into those pores**. The color is in the **oxide walls themselves**, created two ways as the coating grows:

1. **Occluded alloy constituents** — the fine intermetallic particles in the aluminum (Fe, Si, Mn, Cr-bearing) don't fully convert to clear oxide. They get **trapped in the growing film** and **absorb and scatter light**, tinting the coating toward bronze/gray/black.
2. **Organic-acid-modified structure** — the organic acids (sulfophthalic, sulfosalicylic, maleic, oxalic, etc.) change how the oxide grows, producing a **harder, denser, light-scattering film** that pushes the color toward the warm bronze end.

COLOR DEEPENS WITH THICKNESS — SO COLOR = AMP-HOURS

Here is the operator's golden rule of this line: **the thicker you grow the coating, the darker the color**. A thin film is **champagne / light bronze**; grow it thicker and you walk through **medium bronze** → **dark bronze** → **gray** → **black**. Because coating thickness is set by **current density × time** — that is, **amp-hours per unit area** — **color is controlled by amp-hours**. You don't pick a darker dye; you **run more amp-hours** to a deeper color.



Same alloy, same bath — color deepens as the oxide grows thicker (more amp-hours). The color is **in the oxide walls** (with occluded alloy particles), not added to the pores.

APPROX. COATING THICKNESS*	TYPICAL INTEGRAL COLOR (ALLOY-DEPENDENT!)
~5–10 µm	Champagne / light bronze
~10–15 µm	Medium bronze
~15–20 µm	Dark bronze
~20–25 µm	Gray / charcoal
~25–30+ µm	Black

*Representative only. The exact thickness-to-color relationship depends heavily on the alloy/temper and the specific process — **always anodize a sample of the actual metal and run to your shop's color standard.**

**REMEMBER — THE HEART OF INTEGRAL COLOR**

Color is grown, not applied. Darker = thicker = more amp-hours. There is **no dye step** to fix a wrong color. If you stop the run too early, the part is too light; too late, too dark. **Run to amp-hours and to your color standard**, on the correct alloy — that is the job.

**HEADS-UP — THERE'S NO DYE TO RESCUE A MISMATCH**

On a dyed line, a slightly light coating can sometimes be tweaked at the dye tank. **Not here.** The only fix for an out-of-color integral part is to **strip and re-anodize** — expensive and dimension-eating. Get the alloy, amp-hours, and bath right *before* you commit a load.

**TECHNICAL STUFF — WHY HIGHER VOLTAGE AND COLDER**

To build a thick, hard, **deeply colored** film you need more energy than plain decorative anodizing: **higher voltage** (the insulating film grows thick, so volts climb to keep current flowing) and a **well-cooled, agitated bath** so all that energy doesn't overheat and "burn" the coating. This is why integral color sits energetically **between Type II and hardcoat** — sometimes called a "self-coloring hardcoat-type" process. The payoff is hardness + durable color; the price is **electricity and refrigeration**.

**TECHNICAL STUFF — THE ORGANIC ACIDS, BY NAME**

The classic additives are **sulfophthalic acid** (Kalcolor), **sulfosalicylic acid** (Duranodic 300), and others such as **maleic** and **oxalic** acid, used **with** sulfuric acid. They are aromatic/organic acids that toughen and color-shift the oxide. Exact identities and concentrations are **proprietary to each process — read your TDS**. Chemically they are generally **lower-hazard than the chromic acid** used in old chromic anodize (no hexavalent chromium), but they are still **acids and irritants**, and the bath is still **strong sulfuric acid**.

**FUN FACT**

Two extrusions cut from the **same drawing** but made from **different heats of 6063** can come out of an integral-color tank a visibly different bronze at identical amp-hours — because their trace iron and silicon differ. Veteran architectural anodizers obsess over **metal source and lot control** for exactly this reason.

DIMENSIONAL GROWTH

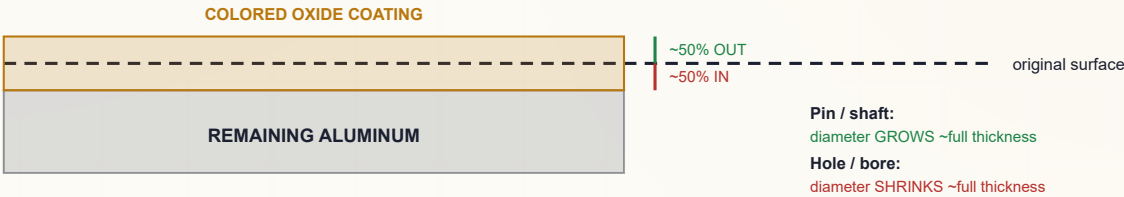
THE COATING MAKES THE PART BIGGER – AND ON A THICK ARCHITECTURAL FILM, YOU MUST PLAN FOR IT

Because we're converting aluminum into aluminum oxide, and oxide takes up **more room** than the metal it came from, the coating grows in **two directions at once**:

- About **half of the coating thickness grows *into*** the original surface (consuming aluminum).
- About **half grows *out*** from the original surface (the part gets bigger).

So for a Class I architectural coating of, say, **20 μm (0.0008")**, the surface moves outward by roughly **10 μm (0.0004") per face**. On a shaft or pin the **diameter grows by about the full coating thickness**; a hole's **diameter shrinks by about the full coating thickness**.

Integral color: ~18–30 μm (Class I) → growth is real



REMEMBER — THE 50% RULE

Anodize grows **roughly half in, half out**. Each *surface* moves out ~half the coating thickness; a *diameter* changes by ~the full coating thickness. Because integral color is run **thick (Class I, $\geq 18 \mu\text{m}$)** to reach its colors, this growth is **not negligible** — tight-fit features must account for it *before* anodizing.



TIP

Architectural extrusions usually have generous tolerances, but **interfaces matter**: snap-fits, screw bosses, weep slots, glazing pockets, and thermal-break channels can be thrown off by a thick coating. When a fabricator asks “how much will it grow?”, the rule of thumb is **~half the spec'd thickness per surface** — confirm against the drawing and your shop standard.



HEADS-UP — MASKING & GROUNDING

Surfaces that must *not* grow or must stay conductive (grounding lugs, screw threads, electrical bonds) are **masked** or machined to allow for growth. Anodize is **electrically insulating** — and an integral-color film is *thicker* than Type II, so a ground path left unmasked will **definitely** not conduct afterward.



TECHNICAL STUFF — PILLING-BEDWORTH

Aluminum oxide occupies more volume than the aluminum it formed from (Pilling–Bedworth ratio > 1). That expansion is *why* the coating grows outward at all — and it's the same expansion that lets hot-water sealing pinch the pores shut later.



FUN FACT

On a long architectural extrusion, the **total growth across a wide face is still just microns** — invisible to the eye — yet it's enough to change a precision fit. The coating you can barely measure can still bind a tight assembly.

CHAPTER 6

THE ANODIZE FAMILY

SAME IDEA, MANY FLAVORS — KNOW WHICH ONE YOUR SPEC IS ASKING FOR

All anodize “types” grow an integral aluminum-oxide coating with the part as the anode. They differ in **acid, thickness, and purpose**.

	TYPE I — CHROMIC	TYPE II — SULFURIC	TYPE III — HARDCOAT	INTEGRAL COLOR (VARIANT)
Electrolyte	Chromic acid	Sulfuric (~165–200 g/L)	Sulfuric, colder/concentrated	Sulfuric + organic acid (sulfophthalic / sulfosalicylic / maleic)
Typical thickness	~0.5–7.5 μm	~5–25 μm	~25–100+ μm	~18–30+ μm (Class I)
Voltage	Lower	~12–18 V	Higher	Higher (~25–75 V)*
Color	Gray, limited	Clear; dyeable	Darkens naturally; can be dyed	Color grown IN the coating — champagne→bronze→black
How color is made	n/a	Dye in pores	Mostly natural/dye	By the anodize itself (no dye)
Main use	Aerospace, fatigue	Decorative + protective	Max wear	Exterior architecture (lightfast color)

*Process-dependent — verify TDS (voltage climbs as the thick film grows).

★

REMEMBER

Type I = chromic, thin, aerospace. Type II = sulfuric, dyeable, the everyday standard. Type III = hardcoat, thick, max wear. Integral color = sulfuric + organic acid, thick, the color is grown in for durable architecture. Integral color is essentially a colored, hardcoat-energy cousin of Type II.

💡

TIP — THE SPECIALTY COUSINS

The other specialty anodizes in this training series: **BSAA** (boric-sulfuric, a chromic replacement for aerospace), **PAA** (phosphoric, for adhesive bonding), **bright anodize** (polished first, mirror look), and the **two coloring cousins** — **integral color** (this manual) and **two-step electrolytic** (next manual). They all stand on the Type II fundamentals.

⚙️

TECHNICAL STUFF — INTEGRAL COLOR VS. HARDCOAT

Both run higher energy than Type II, but with different goals. **Hardcoat (Type III)** maximizes *thickness and hardness* (often run very cold). **Integral color** uses organic acids to maximize *durable color* at architectural thickness. They overlap — integral-color films are notably harder than ordinary Type II — which is part of why integral architectural finishes wear so well.

ALUMINUM ALLOY & TEMPER

ON MOST LINES THE ALLOY AFFECTS CLARITY. HERE, THE ALLOY LITERALLY *MAKES THE COLOR*.

Anodizing only works on **aluminum** (and a few other “valve metals”). On this line it’s aluminum — but on **no other anodize line does the alloy matter as much as it does here**, because the alloy’s own constituents are part of the colorant.

WHY ALLOY & TEMPER RULE THIS PROCESS

The **trace alloying elements** — iron, silicon, manganese, chromium, magnesium — and how they’re distributed (set by **alloy chemistry AND temper/heat treatment**) determine **what color you get at a given thickness**. Change the alloy, the heat, the extrusion press, or the temper, and the **same amp-hours give a different bronze**.

ALLOY / PRODUCT	INTEGRAL-COLOR BEHAVIOR
6063 (“architectural”)	The classic integral-color extrusion alloy; repeatable, attractive bronzes. The workhorse of architectural aluminum.
6061 / other 6xxx	Anodize well but may color differently than 6063 — don’t assume a match.
5xxx sheet (e.g., 5005)	Common for flat sheet/panel ; colors differently from extrusions — a classic matching headache.
2xxx (copper) & high-Si castings	Color dull, gray, mottled, unpredictably — generally not used for quality integral color.

 **REMEMBER — THE MATCHING PROBLEM IS THE PROBLEM**

The central challenge of integral color is **color matching across different mill products and lots**. An **extrusion (6063)** next to a **sheet panel (5005)** next to a **casting** will tend to color **differently at the same settings**, because their constituents differ. Architectural jobs combine all three — so **alloy/temper/lot control and color standards are the whole game**.

 **TIP**

Best practice on every architectural job: **anodize a sample of the actual alloy, temper, and lot** to your **approved color standard** before committing production. Keep extrusions and sheet on **separate, tuned runs** if you can — and **rack like-with-like**. Record the metal source/lot on the traveler.

 **TECHNICAL STUFF — TEMPER MATTERS, NOT JUST ALLOY**

Heat treatment changes the **size and distribution of intermetallic particles**. Two pieces of *the same alloy* in different tempers (e.g., T5 vs T6) can color differently because their particle structure differs. That’s why architectural anodizers specify **alloy AND temper** — and sometimes a specific mill — for color-critical work.

 **FUN FACT**

The aluminum industry developed **special “anodizing-quality” architectural alloys and tight composition controls** specifically so integral color would be repeatable building-to-building. The finish was demanding enough that it shaped how the metal itself was made.

CHAPTER 8

HOW THE LINE WORKS

WALKING THE LINE — EVERY TANK HAS A JOB, AND THE ORDER IS THE RECIPE

An integral-color line is a sequence of tanks. Here's the whole journey and *why each step has to be where it is*.

1. **Inspect / Rack** — Check the part, **confirm the alloy/temper/lot and the color standard**, and clamp it tight. The part **is the anode**, so contact is everything; a loose part = no coating or a burned part. The contact point won't anodize, so it's chosen carefully.
2. **Clean / Degrease** — Remove oils and soil; oxide won't grow evenly through grease.
3. **Etch (caustic)** — A hot caustic dip removes the natural oxide skin and gives the **architectural satin/matte finish** most building aluminum has. It also evens the surface — important for **uniform color**.
4. **Rinse** — Wash off caustic before the acid steps.
5. **Desmut / Deox** — Remove the dark **smut** (alloy constituents) the etch leaves, for a clean, uniform start. *(The smut here is made of the very elements that will color the coating — so consistent desmut helps consistent color.)*
6. **Rinse** — Clean before the main tank.
7. **COLOR-ANODIZE** — The main event. Part = anode, **sulfuric + organic acid**, well cooled, **higher voltage**, run to **amp-hours / color**. The colored oxide grows here.
8. **Rinse** — Gently rinse acid out of the fresh oxide before sealing.
9. **(No Dye)** — There is **no dye step** — the color is already in the coating. *(This tank position is skipped or used only as a rinse/hold.)*
10. **Seal** — Close the pores for corrosion and abrasion resistance (the color is already locked in).
11. **Rinse / Dry** — Final clean-up and gentle dry.
12. **Inspect** — **Color/ΔE vs. standard**, thickness, coating weight, seal quality.



REMEMBER — WHY THE ORDER CAN'T MOVE

You can't anodize over grease or smut (uneven, off-color oxide). You **grow the color in the anodize tank** — so there's nothing to "add" afterward. You can't seal before the color is fully grown. The line *is* the recipe.



HEADS-UP — THE LINE IS A TOUR OF HAZARDS

As you walk the line you pass **hot caustic** (etch), **strong sulfuric + organic acids** (desmut, color-anodize), **flammable hydrogen** (cathode), **higher-voltage DC** (rectifier/contacts — more shock/arc energy than a Type II line), and **near-boiling water** (seal). Each station chapter calls out its own. Respect each tank for what *it* is.



TIP

When learning, walk the empty line with your trainer and name each tank's *job* and *main hazard* out loud — and add one integral-color-specific line: **"the color is made at Station 07, controlled by amp-hours and the alloy."**

CHAPTER 9

PPE — YOUR DAILY ARMOR

WHAT YOU WEAR, AND WHY EACH PIECE IS THERE

An integral-color line works with strong sulfuric **plus organic acids**, hot caustic etch, a near-boiling seal, **higher-current/higher-voltage DC**, and chemical mist. None of these will hurt you if you respect them and wear the right gear. All of them can hurt you badly if you don't.

HAZARD ON THE LINE	PPE THAT PROTECTS YOU
Sulfuric + organic acids (anodize, desmut) — burns, splash, mist	Sealed chemical splash goggles + face shield , chemical-resistant gloves , apron/suit , chemical-resistant boots
Hot caustic etch — burns worse than you'd expect	Same as above; caustic burns are deep and slow to feel at first
Near-boiling seal tank (~96–100 °C) — scald/steam	Face shield, heat-resistant gloves, long sleeves; mind the steam
Acid mist / fumes	Local exhaust at tanks; respiratory protection per your program
Higher-voltage DC / arcing at contacts	Dry hands, dry gloves, no jewelry, follow LOTO — more stored energy here than a Type II tank
Handling organic-acid powders during make-up	Dust protection per SDS; avoid inhalation; some organics are combustible as dust
Slips on wet floors	Non-slip chemical-resistant boots

 **TIP**

Put your gear on *in order*: boots and apron first, then gloves, then goggles, then face shield last. Take it off in reverse so you never touch your face with contaminated gloves. Rinse your gloves *before* you take them off.

 **REMEMBER**

On an anodize line your **eyes and skin** are the most exposed. **Sealed splash goggles and a face shield are not optional** at the working tanks.

 **HEADS-UP — GOGGLES, NOT JUST GLASSES**

Safety glasses stop flying chips; they do **not** stop an acid or caustic splash from the side. At every chemical tank, sealed splash goggles (and a face shield when charging, dumping, or handling parts) are required.

 **HEADS-UP — NO EATING/DRINKING/VAPING ON THE LINE**

Hand-to-mouth contact is a major route for ingesting chemicals. Wash thoroughly before breaks and before leaving. Keep food and drink completely out of the work area.

CHAPTER 10

EMERGENCY & FIRST-RESPONSE

KNOW THESE COLD BEFORE YOU TOUCH A TANK

Know these *before* you need them. In an emergency, seconds matter and you won't have time to read.

SITUATION	WHAT TO DO – IMMEDIATELY
Acid or caustic on skin	Flush immediately with water for at least 15 minutes . Remove contaminated clothing while flushing. Get medical attention — caustic burns can be deep even when they don't hurt at first.
Acid or caustic in eyes	Go straight to the eyewash , hold eyes open, flush at least 15 minutes , get medical help. Don't stop early.
Large splash / drench	Use the safety shower — full 15 minutes, clothing off.
Hot seal-tank scald	Cool the burn with water; don't pop blisters; get medical attention for anything beyond minor.
Acid spill	Don't play hero — alert others, contain only if trained, use the spill kit, follow your shop's plan. Never mix acids and caustics during cleanup.
Electrical / arc at rectifier or contacts	Don't touch — de-energize via rectifier controls / LOTO. Never reach into a tank with current on. Higher voltage here = more arc energy.
Hydrogen gas	Keep ignition sources (flames, sparks, grinding) away from the cathode/tank area; ensure ventilation is running.
Fluoride exposure (if a fluoride desmut/seal is used)	Fluoride burns are deep, delayed, and systemically toxic — flush, then apply the specific first-aid in the SDS (often calcium gluconate gel) and get medical help immediately. Ordinary flushing alone may not be enough.

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HEADS-UP — THE GOLDEN RULE FOR ACID MAKE-UP
Always add **ACID to WATER**, never **water to acid**. (“Do as you *oughta*: add acid to water.”) Adding water to concentrated sulfuric can flash-boil and erupt. Bath make-up is usually authorized-personnel work — but every operator must know this rule. The **same rule applies to the organic acids**.

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HEADS-UP — LOCKOUT/TAGOUT (LOTO)
LOTO is mandatory for *all* rectifier and electrical maintenance. Physically lock the power off and tag it. On a higher-voltage integral-color rectifier, the stored energy is greater — “I’ll just be a second” is how people get electrocuted.

★

REMEMBER
Know where the **nearest eyewash, safety shower, and spill kit** are at every station before you start your shift. Eyewash and shower stations must always be unblocked.

CHAPTER 11

THE SAFETY PHILOSOPHY OF AN INTEGRAL-COLOR LINE

THE MINDSET THAT TIES EVERY STATION TOGETHER

1. TWO CORROSIVES, MIRROR-IMAGE.

The **etch** is **caustic** (high pH); the **desmut** and **color-anodize** are **acid** (low pH). Both burn. Treat high pH and low pH with equal respect — and never let them mix outside a controlled neutralization.

2. THE PART IS LIVE — AND AT HIGHER VOLTAGE.

Because the part is the anode, there is **DC current at the work** and at the contacts — and integral color runs **higher voltage** than Type II, so there's **more electrical energy** to respect. Wet hands + current + acid is a bad combination.

3. HOT IS A HAZARD TOO — AND THERE'S MORE HEAT HERE.

The seal tank runs **near boiling**, and the color-anodize bath **self-heats hard** because of the high energy input. Steam scalds; a hot-acid splash is as dangerous as a chemical one.

4. HYDROGEN IS FLAMMABLE.

The cathode bubbles off hydrogen gas. Ventilation on, ignition sources away.

5. ENGINEERING CONTROLS FIRST, PPE SECOND.

Tank exhaust, **robust cooling**, lids, and splash guards do the heavy lifting; PPE is your last layer, not your first. And when in doubt, **stop and ask** — the unsafe operator isn't the one who asks questions; it's the one who guesses.

• THE HAZARD FAMILIES TO INTERNALIZE

1. **Corrosives (acid AND caustic).** Sulfuric + organic acids (anodize/desmut) and caustic (etch). *Defense:* goggles, face shield, gloves, apron; 15-minute flush; acid-to-water.
2. **Hot / thermal.** Near-boiling seal, hot etch, and a **high-energy, self-heating anodize bath**. *Defense:* face shield, heat gloves, steam awareness, **watch the cooling**.
3. **Electrical & gas. Higher-voltage DC;** flammable hydrogen at the cathode. *Defense:* LOTO, dry hands, no ignition sources, ventilation.



REMEMBER

Corrosives first, thermal second, electrical/gas third. If you can name the hazard family at the tank in front of you, you already know most of what you need to protect yourself. The integral-color line just turns the **thermal and electrical dials up** — so respect those two a little more than on a Type II line.



FUN FACT

Honest framing: integral color is **safer than the old chromic-acid color routes** (no hexavalent chromium) and the organic acids are generally **lower-hazard than chromic** — but it is **not a “green,” low-energy process**. It trades chemical hazard down and energy/heat hazard up. Know which hazards you actually have.

PART TWO — STATION BY STATION
STATION 01 OF 12

INSPECTION & RACKING

“THE PART IS THE WIRE. A LOOSE PART IS A DEAD PART — AND HERE, THE WRONG ALLOY IS A WRONG COLOR.”

PURPOSE & THE “WHY”

Inspect the incoming aluminum, **confirm the alloy, temper, lot, and the approved color standard** (integral color, target thickness/color, AAMA 611 class, sealed), and **rack it** for firm current-carrying contact. Because the part **is the anode**, every bit of current that grows your colored coating flows *through the rack contact*. A loose or corroded contact means high resistance — **light/no color, a burned spot, or a dropped part**. Racking also decides *where* the un-anodized contact mark lands and how gas/solution drains.



REMEMBER — ALLOY/LOT IS PART OF INSPECTION HERE

On most lines, racking is about contact. On *this* line, racking starts with **confirming the metal** — because **the alloy and temper set the color**. Rack **like-alloy with like-alloy** and keep extrusions separate from sheet, or you'll get a multi-shade load. Record the alloy/temper/lot on the traveler.



REMEMBER — THE CONTACT MARK IS UNAVOIDABLE

Wherever the rack grips, **no oxide grows** (that spot must carry current). Place the contact where the bare mark won't show — a hidden face, a concealed end, a fastener hole — and make it **tight**.

ITEM	SPEC / PRACTICE	WHAT TO WATCH
Incoming inspection	Per traveler / architectural spec	Confirm alloy, temper, lot , color standard, thickness/class
Rack material	Titanium or aluminum	Ti racks don't anodize away; Al racks anodize and need stripping
Contact pressure	Firm, spring-loaded / bolted	Loose = burning, light color, dropped parts
Contact location	Hidden/non-critical area	The contact point won't anodize (bare mark)
Masking	Plugs/tape/lacquer on no-anodize areas	Grounds, threads, concealed fits
Orientation	Allow gas escape & drainage	Avoid trapped air (un-anodized, off-color voids)
Like-with-like	Group by alloy/temper/lot	Mixed metals = mixed color in one load



HEADS-UP

Racks and parts have **sharp edges and pinch points**, handled near chemical tanks. A part that drops from a weak rack becomes a **splash hazard** in an acid or caustic tank, not just scrap.



TIP — TITANIUM RACKS

Titanium racks don't anodize (they passivate), so they hold contact load after load. Aluminum racks anodize along with the part and must be **stripped/re-pointed** regularly.

STEP-BY-STEP

1. Read the traveler (**alloy, temper, lot**, color standard, thickness/class, masked areas). 2. Inspect for scratches/gouges/die lines — anodize *reveals* flaws. 3. Plan a hidden contact that carries full current. 4. Mask no-anodize areas. 5. Rack firmly, correct orientation, **like-alloy together**. 6. Verify no parts touch (shadowing). 7. Handle by the rack from here on.

SIGN-OFF — STATION 01

Teach-Back: Why contact matters more in anodizing (part = anode); why the contact point won't anodize; **why alloy/temper/lot must be confirmed and grouped here (it sets the color)**; why titanium racks are preferred; two no-anodize areas that get masked and why.

“I verified alloy/temper/lot and the color standard, grouped like-with-like, made a firm hidden contact, masked all no-anodize features, racked for gas escape and drainage with no parts touching, and handled parts by the rack.”

Trainee: _____ Trainer: _____ Date: _____

STATION 02 OF 12

CLEAN / DEGREASE

“ANODIZE OVER OIL AND YOU’VE GROWN A DEFECT, NOT A COATING – AND A BLOTCHY COLOR.”

PURPOSE & THE “WHY”

Get the part **chemically clean** — no oil, grease, fingerprints, or soil — so the colored oxide grows **evenly**. Oil and soil cause **uneven anodizing, streaks, and patchy color** — and on an integral-color line, *patchy oxide means patchy color* with no dye to mask it. Cleaning is usually a **mild non-etching alkaline soak cleaner** (formulated *not* to attack the aluminum — we save controlled metal removal for the etch). The free, foolproof check is the **water-break test**.



TIP — THE WATER-BREAK TEST (FREE, FAST, NEVER LIES)

Pull a cleaned part and watch the water. A truly clean surface sheets water in **one continuous film** (PASS). A dirty surface makes water **bead up** (FAIL — re-clean). Costs nothing; takes two seconds; use it on every part.

FAIL - water beads up
+-----+
| o o o | X
| o o |
+-----+
Oil remains -> RE-CLEAN

PASS - water sheets evenly
+-----+
|#####| OK
|#####|
+-----+
Surface clean -> proceed

PARAMETER	RANGE	WHAT TO WATCH
Type	Mild non-etching alkaline cleaner	Must not attack aluminum
Temperature	~50–70 °C (120–160 °F)	Per cleaner TDS
Time	~3–10 min	Heavy oil longer
Concentration	Per TDS	Titrate regularly
pH	Mildly alkaline	Still a skin irritant/burn risk
Water-break test	Pass = continuous sheet	The only reliable field check



HEADS-UP — HOT ALKALINE SOLUTION

Even a “mild” cleaner is hot and alkaline: splashes irritate/burn skin and eyes; steam irritates airways. Splash goggles, gloves, apron. Skin → flush 15 min; eyes → eyewash 15 min and get help.

STEP-BY-STEP

1. Check temperature and concentration. 2. Immerse the racked load fully; start the timer; confirm agitation. 3. Withdraw slowly, draining back. 4. Run the **water-break test** — sheet → proceed; beading → re-clean. 5. Move promptly to etch.

SIGN-OFF — STATION 02

Teach-Back: What the cleaner removes (oil/soil) vs. what it must *not* do (attack aluminum — that’s the etch); perform/read a water-break test; **why patchy cleaning shows up directly as patchy color here**; the caustic/irritant hazard and PPE.

“The part passed the water-break test, the cleaner was in range and at temperature, and I wore required PPE.”

Trainee: _____ Trainer: _____ Date: _____

STATION 03 OF 12

ETCH (ARCHITECTURAL SATIN)

“WHERE THE PART GETS ITS ARCHITECTURAL SATIN LOOK – AND WHERE YOU’LL RESPECT CAUSTIC FOREVER.”

PURPOSE & THE “WHY”

Remove the natural oxide skin and a thin layer of aluminum in a **hot caustic (sodium hydroxide) etch**, giving the uniform, matte **satin/matte finish** that defines architectural aluminum, and a fresh, even surface for color-anodizing. On architectural work the etch is often run to a **deliberate satin or matte** — the look on most building extrusions. Crucially, an **even etch means even color**: an uneven or streaky etch will translate into **streaky bronze**. Etch too little → shiny, blotchy; etch too much → lost dimension/detail.



REMEMBER — ETCH SETS TEXTURE AND COLOR UNIFORMITY

The etch decides **satin vs. matte** and contributes to **color uniformity**. Architectural integral color almost always starts from a **uniform caustic satin etch**. Inconsistent etch time, temperature, or concentration across a load = visible color variation later — with no dye to hide it.

PARAMETER	RANGE	WHAT IT DOES / WATCH
Chemistry	Sodium hydroxide, often with additives	Dissolves oxide + aluminum
Concentration	~40–60 g/L NaOH	Higher = faster/more aggressive
Temperature	~50–65 °C (120–150 °F)	Hot — burns and speeds etch
Time	~1–10 min (per architectural finish)	Longer = more matte + more metal loss
Result	Uniform satin + smut	Smut is normal — removed at desmut



HEADS-UP — HOT CAUSTIC IS ONE OF THE WORST BURNS ON THE LINE

Caustic burns are **deep, get worse with time, and may not hurt at first**. Full PPE: sealed goggles, face shield, elbow-length chemical gloves, apron, boots. Skin → flush **15+ minutes**; eyes → eyewash 15 min and get help. Never assume “it’s just a little.”



HEADS-UP — CAUSTIC + ALUMINUM MAKES HYDROGEN

Caustic attacking aluminum releases **hydrogen gas** (flammable). Keep ignition sources away and ventilation running.

STEP-BY-STEP

1. Confirm concentration and temperature. 2. Immerse; time the **specified etch** (controls finish, color uniformity, and metal loss). 3. Agitate per procedure. 4. Withdraw and drain — part will look **dark/smutty** (normal). 5. Move **directly to rinse** — don’t let caustic dry on the part.

SIGN-OFF — STATION 03

Teach-Back: What the etch removes and the finish it creates (architectural satin); **why etch uniformity drives color uniformity**; why the part comes out dark/smutty and what fixes it (desmut); the caustic-burn hazard and first response.

“I etched for the specified time/concentration/temperature for a uniform satin, understood the color-uniformity and dimensional effects, moved straight to rinse, and wore full PPE for hot caustic.”

Trainee: _____ Trainer: _____ Date: _____

STATION 04 OF 12

RINSE (THE FIREWALL)

“A RINSE ISN’T A REST STOP. IT’S A CHECKPOINT.”

PURPOSE & THE “WHY”

Wash the **caustic film** off the part before it reaches the acidic desmut and color-anodize chemistry. The etch is caustic (high pH); desmut and anodize are acidic (low pH). Carry caustic drag-out forward and it **neutralizes and contaminates the acid baths**, wastes chemistry, and causes staining/color variation. Cleanliness is tracked by **conductivity** (µS/cm) — *lower = cleaner*.



TIP — COUNTER-FLOW RINSING

The professional approach is a **counter-current** cascade: parts move forward, water flows backward, so the cleanest water touches the part last. It saves enormous water — important on a line with this many rinses.

PARAMETER	RANGE	NOTES
Temperature	Ambient	No heating needed
Time	30–60 s with agitation	Longer for cavities/blind holes
Water source	Municipal or DI	DI preferred near anodize/seal
Conductivity	< 500 µS/cm (target < 300)	Monitor at shift start
Flow	Continuous overflow / counter-current	Prevents stagnation



HEADS-UP

This rinse carries dragged-in caustic and starts mildly alkaline. Goggles, gloves, non-slip footwear. **Wet floors and slips are the #1 everyday injury at rinses.**

DO

- Check conductivity at shift start; tell your lead if near/above limit.
- Transfer promptly from the etch — don’t let caustic dry.
- Immerse fully with agitation 30–60 s (longer for cavities).
- Lift and drain over the rinse tank before moving on.

TOP MISTAKES

- Treating the rinse as a quick dunk.
- Ignoring conductivity.
- Letting parts drip-dry between etch and rinse.
- Not flushing cavities; skipping the drain step.

SIGN-OFF — STATION 04

Teach-Back: Why carrying caustic into the acid stages causes problems; define drag-out / drag-in; what conductivity tells you and which way is “clean.”

“Rinse conductivity was within limit, water flow was running, and parts were rinsed the full time with PPE worn.”

Trainee: _____ Trainer: _____ Conductivity: _____ µS/cm

STATION 05 OF 12

DESMUT / DEOX

“ETCHING UNCOVERS THE ALLOY’S LEFTOVERS. DESMUT WASHES THEM AWAY — CONSISTENTLY, FOR CONSISTENT COLOR.”

PURPOSE & THE “WHY”

Remove the dark **smut** the etch left — the alloying elements (copper, silicon, iron, manganese) that didn’t dissolve in caustic — using an **acid desmut/deox**, leaving a bright, uniform surface for color-anodizing. Anodize over smut and you get **dark streaks, dull patches, and uneven color**. On an integral-color line there’s an extra wrinkle: those alloy constituents are the **same elements that color the coating**, so **consistent desmut helps consistent color**. Desmut chemistry depends on the alloy: simple alloys may need only **nitric** or **sulfuric**; high-silicon/copper alloys often need a **fluoride-bearing (or proprietary) deox**.



REMEMBER

Smut after etch is normal. Desmut’s job is to remove it uniformly. A part entering the color-anodize tank still gray/smitty will come out **streaky and off-color** — and there’s no dye to cover it.

PARAMETER	RANGE / PRACTICE	NOTES
Chemistry	Nitric and/or sulfuric; fluoride-bearing/proprietary for high-Si/Cu	Match to the alloy
Concentration	Per product TDS	Titrate regularly
Temperature	Often ambient-warm	Per TDS
Time	~0.5–3 min	Until smut is gone, no longer
Result	Bright, uniform, active surface	No gray film remaining



HEADS-UP — STRONG ACID (AND POSSIBLY FLUORIDE)

Desmut is acidic; **fluoride-bearing desmuts are especially hazardous** (HF-type chemistry — deep, delayed burns; systemically toxic). Know which chemistry your tank uses, read its SDS, wear full acid PPE, and know the **specific first-aid** (fluoride burns may need **calcium gluconate**). **Add acid to water** for make-up.

STEP-BY-STEP

1. Confirm which desmut chemistry (and its hazards/first-aid). 2. Check concentration/temperature. 3. Immerse briefly — just until smut clears. 4. Withdraw, drain, rinse. 5. Don’t over-dip — excessive desmut can pit some alloys.

SIGN-OFF — STATION 05

Teach-Back: What smut is and where it came from; why anodizing over smut causes defects; **why consistent desmut supports consistent color here**; that desmut chemistry depends on the alloy; the fluoride hazard/first-aid if applicable.

“I removed the smut uniformly with the correct desmut for the alloy, did not over-dip, and understand the specific hazard and first-aid for this tank.”

Trainee: _____ Trainer: _____ Chemistry: _____

STATION 06 OF 12

RINSE (PRE-ANODIZE)

“THE LAST CLEAN HANDOFF BEFORE THE MAIN EVENT.”

PURPOSE & THE “WHY”

Remove desmut acid before the color-anodize tank. Carrying desmut drag-out introduces unwanted ions (and, with a fluoride desmut, **fluoride contamination** that can roughen/pit the anodize and disturb color). A clean rinse protects the bath.

PARAMETER	RANGE	NOTES
Temperature	Ambient	—
Time	30-60 s, agitation	Longer for cavities
Water	DI preferred	Cleaner handoff
Conductivity	Per shop spec	Lower = cleaner
Flow	Counter-current overflow	—



HEADS-UP

Same rinse rules: drag-in acid, wet floors, goggles/gloves. If the prior tank was a **fluoride** desmut, this rinse carries fluoride — handle and dispose accordingly.



TIP

Move into the color-anodize tank **promptly and consistently**. A long, variable dwell before anodizing causes part-to-part **thickness and therefore color** differences. On this line, **consistency of timing = consistency of color**.

STEP-BY-STEP

- Confirm flow and conductivity.
- Rinse the full time with agitation.
- Drain over the tank.
- Move straight to anodize.

TOP MISTAKES

- Quick dunk instead of a real rinse.
- Inconsistent dwell before anodize.
- Ignoring conductivity.
- Forgetting fluoride drag-in handling.

SIGN-OFF — STATION 06

Teach-Back: Why a clean handoff to the anodize tank matters; **why a consistent dwell before anodize protects color**; what fluoride drag-in could do.

“The part was fully rinsed, conductivity was acceptable, and I moved promptly and consistently to the color-anodize tank.”

Trainee: _____ Trainer: _____ Date: _____

STATION 07 OF 12 • THE MAIN EVENT

COLOR-ANODIZE — THE MAIN TANK

WHERE THE ALUMINUM’S OWN SURFACE BECOMES A HARD, COLORED CERAMIC — AND WHERE THE COLOR IS BORN

Take a breath. Up to now, every station has had one job: get the part clean, even, and ready. **Station 07 is where the color is actually created** — the tank that defines this whole line.

The big idea in one sentence: **we run the aluminum part as the *anode* in a cooled, sulfuric-plus-organic-acid electrolyte and use higher-voltage DC to grow a hard, *colored* aluminum-oxide layer out of the part’s own surface — and the color deepens the longer/harder we run.** The cast:

- **Anode** — *your aluminum part (positive)*. Where the colored oxide grows.
- **Cathode** — **lead or aluminum (6063) plates** (negative). Carry current, bubble off hydrogen; they do **not** feed metal or color into the part.
- **Electrolyte** — **dilute sulfuric acid blended with one or more organic acids** (sulfophthalic / sulfosalicylic / maleic / oxalic — proprietary), held cool with strong cooling and agitation.

★

REMEMBER — THE COLOR IS GROWN HERE, CONTROLLED BY AMP-HOURS

The cathodes don’t supply the color — **the color comes from the part’s own alloy plus the organic-acid-modified oxide**, and it gets darker the more amp-hours you run. There is no later dye step. **This tank IS the color step.**

⚠

HEADS-UP — SULFURIC + ORGANIC ACIDS + HIGHER-VOLTAGE DC + HYDROGEN + A BIG HEAT LOAD, ALL AT ONCE

This tank holds **strong acids** (burns, splash, mist), runs **higher-voltage DC** than a Type II tank (more shock/arc energy at contacts), bubbles **flammable hydrogen** off the cathodes, and **self-heats hard** because of the high energy input (cooling must work or the coating burns and color drifts). Engineering controls — **exhaust, robust cooling, agitation, splash control** — must be working before you run it. Full acid PPE. No ignition sources.

• WHY INTEGRAL COLOR (AND WHAT MAKES IT PARTICULAR)

ADVANTAGE	WHAT IT MEANS ON THE FLOOR
Color grown into the oxide	No dye; extremely lightfast
Hard, dense, abrasion-resistant film	Survives weather, cleaning, handling
Corrosion resistance (sealed)	Decades of exterior service
Architectural Class I capable	Meets AAMA 611 / ASTM B580 thickness

...but the same chemistry demands care: it must be **kept cool** (warm bath = soft/powdery/burned + off-color), it needs **steady current/amp-hours** (color depends on it), it’s **alloy-sensitive** (color varies with the metal), it’s **energy-intensive**, and it grows the coating **half into the part** (dimensional growth).

• THE BATH MAKE-UP — KNOW THE RANGES



HEADS-UP

Integral-color parameters are **process-specific**. The ranges below are representative for organic-acid-modified sulfuric processes (Kalcolor, Duranodic, and modern equivalents) — **verify against your specific process TDS**. The proprietary processes differ enough that a single hard number would mislead.

COMPONENT / PARAMETER	TYPICAL RANGE (VERIFY TDS)	WHAT IT DOES / WATCH
Sulfuric acid (H ₂ SO ₄)	~100–175 g/L	Primary electrolyte
Organic acid (sulfophthalic / sulfosalicylic / maleic)	Per process (proprietary)	Modifies oxide → hardness + color
Temperature	Cool, tightly controlled (often ~18–27 °C)	High energy = big heat load; warm = burned/off-color
Voltage (DC)	~25–75 V (climbs as film grows)	Higher than Type II; drives the colored film
Current density	~18–36 ASF (process-dependent)	Sets growth/color rate
Coating thickness	~18–30+ μm (Class I ≥18 μm)	Sets the color; controlled by amp-hours
Amp-hours	Run to amp-hours / color standard	The real color control variable
Time	~30–60+ min (to target color)	Longer = thicker = darker
Cathodes	Lead or aluminum 6063	Carry current; bubble H ₂ ; don't feed coating
Agitation	Vigorous air agitation	Even temperature & color
Cooling	Strong refrigeration / heat exchanger	Bath self-heats — <i>must</i> be cooled
Dissolved aluminum	Per spec	Too much dulls color & slows the bath
Chloride contamination	Very low	Causes pitting/burning



REMEMBER — THE NUMBERS TO CARRY

Sulfuric + organic acid, cool & tightly controlled, higher voltage (~25–75 V), run to AMP-HOURS / COLOR, ~18–30 μm (Class I), strong cooling + agitation, lead/6063 cathodes. And the one that's unique to this line: **color = amp-hours = thickness, on the correct alloy**.



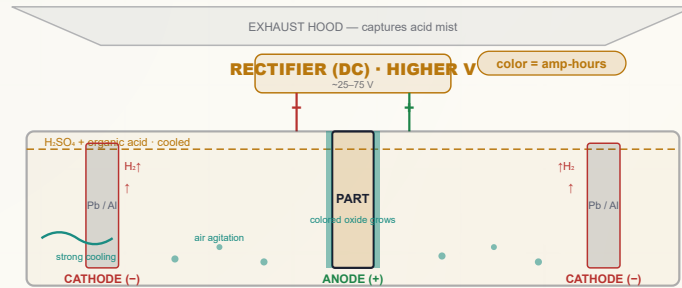
TECHNICAL STUFF — COLOR IS AMP-HOURS, NOT JUST MINUTES

Thickness (and therefore color) is set by **charge passed per unit area = current × time ÷ area = amp-hours/ft²** (or A·min/dm²). That's why integral-color shops **run to an amp-hour count and to a physical color standard**, not just a stopwatch — a part with a poor contact draws less current, accumulates fewer amp-hours, and comes out **too light**. Faraday's law in a hard hat: **more charge, thicker oxide, darker color**.



TECHNICAL STUFF — TEMPERATURE & CONTAMINATION

A **warm bath** dissolves the growing oxide too fast → soft, powdery, “burned,” off-color coating; integral color dumps **more energy** than Type II, so the cooling load is larger and a cooling failure ruins color faster. **Chloride** (tap water, sweat, drag-in) is the silent killer — even tens of ppm cause **pitting and burning**. Use DI water; a little dissolved aluminum is normal, but **too much grays the color**.



Key params: H₂SO₄ + organic acid • cool/controlled • ~25-75 V • run to AMP-HOURS/COLOR • 18-30um • strong cooling + agitation. Note the **reversed polarity** vs. a plating cell: the part is the **ANODE (+)**.

• STANDARD OPERATING PROCEDURE

1. Confirm the part is clean, etched, desmutted, rinsed, and racked tight — straight from the rinse, **consistent dwell** (timing affects color).
2. Verify engineering controls **FIRST**: cooling running and at temperature, agitation on, exhaust on, splash control in place, full acid PPE on.
3. Check bath readiness: sulfuric + organic acid in range (per lab), temperature in range, dissolved-aluminum and chloride within limits, cathodes sound.
4. Confirm the alloy/temper/lot matches the color standard for this run — *the metal sets the color*.
5. Confirm rectifier polarity: **PART = ANODE (POSITIVE)**. (*Opposite of plating — verify every time.*)
6. Set current for the part's surface area to hit target current density.
7. Set the **AMP-HOUR target** (not just a clock time) for the target thickness/color, per your color standard.
8. Energize and ramp per procedure; watch **voltage rise** as the thick insulating colored oxide grows (normal — and it rises higher than on Type II).
9. Anodize to the amp-hour/color target, monitoring temperature, current/voltage, agitation, and cooling the whole run.
10. At target, stop current, withdraw, drain, and move to post-anodize rinse. The part is **already its final color**.
11. Log everything: amp-hours, current/voltage, temperature, alloy/lot, color check, additions.



HEADS-UP — VERIFY POLARITY EVERY SINGLE TIME

Because everyone trained on plating expects the part to be the **cathode**, the most dangerous habit on this line is assuming polarity. **The part is the ANODE (+)**. Reversed polarity gives no coating (and can damage parts/cathodes). Check before you energize.

COMMON DEFECTS — WHAT YOU’LL SEE AND WHAT IT MEANS

WHAT YOU SEE	MOST LIKELY CAUSE	WHAT TO DO
Color too light	Too few amp-hours; poor contact; low CD; bath too warm	Run to amp-hours; re-rack tight ; check CD/temperature
Color too dark	Too many amp-hours; bath off-spec	Reduce amp-hours; verify bath
Color won’t match part-to-part	Mixed alloys/temper/lots ; inconsistent dwell/temp/amp-hours	Rack like-with-like ; standardize timing & amp-hours; sample first
Extrusion vs. sheet mismatch	Different mill products color differently	Separate, tuned runs; match to standard, not to each other blindly
Powdery, soft, chalky, burned	Bath too warm; CD too high; cooling failing	Restore cooling/agitation; lower CD; verify temperature
Thin / no coating (and too light)	Bad contact; wrong polarity; low current; low acid	Re-rack; verify part = anode ; check current/acid
Pitting	Chloride contamination ; impurities	Lab-check chloride; use DI; treat bath
Dull / muddy color	High dissolved Al; off-spec organic acid; high-Cu/Si metal	Lab-check bath; confirm alloy
Streaks	Uneven etch/desmut; smut left on; uneven agitation	Verify clean/etch/desmut; check agitation
Voltage spikes / won’t hold current	Poor contact; failing cathodes; low conductivity	Check contacts/cathodes; lab-check bath



REMEMBER — FOUR ROOTS OF DEFECTS

Most integral-color defects trace to one of four roots: **amp-hours/contact** (too light/dark, banding), **temperature/cooling** (burning), **alloy/lot** (mismatch, muddy), or **contamination/seal** (pitting, bloom). Name the root, not just the symptom — and remember there is **no dye to rescue** an out-of-color part; the fix is strip & re-anodize.



TIP — SAMPLE BEFORE YOU COMMIT A LOAD

On any color-critical or new lot, **anodize a coupon of the actual alloy/temper/lot** to your approved color standard *first*. Confirming amp-hours-to-color on the real metal is far cheaper than stripping a full production load that came out the wrong shade.

INTEGRATED SAFETY — RESPECT THE ENERGY AND THE ACID



HEADS-UP — SULFURIC + ORGANIC ACIDS (THE HEADLINE)

Corrosive; splash and **acid mist** hazard. Engineering controls first (exhaust, splash guards), PPE always. Any contact → flush **15 minutes**, get medical attention. **Acid to water** for make-up — including the organic acids.



HEADS-UP — HIGHER-VOLTAGE DC & FLAMMABLE HYDROGEN

Integral color runs **higher voltage** than Type II — **more shock/arc energy**. LOTO for all electrical work; dry hands only. Hydrogen off the cathodes is **flammable** — ventilation on; no flames, sparks, or grinding nearby.



HEADS-UP — WATCH THE COOLING THE WHOLE RUN

Runs are long and the bath self-heats **harder** than a Type II tank. If cooling fails or temperature climbs mid-run, the coating **burns/powders** and the **color drifts**. Don't set-and-forget.

TEACH-BACK — CAN YOU ANSWER?

- Which electrode is your part? (*Anode, +.*)
- Where do the coating and color come from? (*Part's own Al + organic-acid oxide — not the cathodes.*)
- How is **color controlled**? (*Amp-hours/thickness on the right alloy.*)
- Why must the bath stay cool? What does chloride do?
- The headline numbers (acid, V, μ m)?
- The three safety families (acid, higher-V DC, hydrogen)?

TOP MISTAKES TO AVOID

- Assuming the part is the cathode (it's the **anode**).
- Running by clock instead of **amp-hours**.
- Loose contact (light color / burn).
- Mixing alloys/lots on a rack.
- Ignoring chloride / dissolved-Al; running with failing cooling.
- Eating/drinking near the acid tank.

SIGN-OFF — STATION 07

*"I can independently verify cooling/agitation/exhaust/PPE; verify bath readiness (acid + organic acid, temp, dissolved Al, chloride per lab); confirm the part is racked as the ANODE (positive) and that the alloy matches the color standard; **run to amp-hours/color**; recognize light/dark/burn/pitting/streak defects; and state the acid, higher-voltage DC, and hydrogen hazards. I understand chemistry adjustments are made only by authorized personnel."*

Operator: _____ Date: _____ Trainer: _____ Date: _____

STATION 08 OF 12

RINSE (POST-ANODIZE)

“GENTLE NOW – THE PORES ARE WIDE OPEN, AND THE COLOR IS ALREADY IN.”

PURPOSE & THE “WHY”

Rinse the acid out of the freshly anodized part **without contaminating it**, before sealing. The fresh oxide is **porous and absorbent**. Leftover acid interferes with sealing. Use **clean DI** and keep it **cool** before sealing — warm water here can begin to seal the pores early and unevenly. (There’s no dye step to protect, but a clean, consistent rinse still protects the **seal** and the **final appearance**.)



REMEMBER

After color-anodizing, **use cool DI rinse before sealing**. The color is already locked into the oxide — but a dirty or hot rinse here can still cause **uneven seal and surface blemish** on an otherwise perfect colored part.

PARAMETER	RANGE	NOTES
Water	DI strongly preferred	Pores absorb impurities
Temperature	Cool / ambient (before seal)	Avoid premature/uneven sealing
Time	1-3 min, gentle agitation	Flush acid from pores
Conductivity	Low (per spec)	Confirms acid removed



HEADS-UP

This rinse carries dragged-out **sulfuric + organic acid** — mildly acidic, slip hazard, goggles/gloves. Don’t let acid-laden rinse go untreated to drain.

STEP-BY-STEP

- Confirm DI flow, cool temperature.
- Rinse gently, full time.
- Drain.
- Move to seal.

TOP MISTAKES

- Hot water before seal (uneven seal).
- Hard tap water (impurities).
- Short rinse leaving acid behind.

SIGN-OFF — STATION 08

Teach-Back: Why the pores are absorbent now; why cool DI before seal; that the color is already set (no dye to protect) but appearance/seal still depend on a clean rinse.

“I rinsed with cool DI to flush acid from the open pores without pre-sealing, and moved promptly to seal.”

Trainee: _____ Trainer: _____ Date: _____

— STATION 09 OF 12 • INTENTIONALLY EMPTY

DYE (NOT USED)

“THE COLOR IS ALREADY IN THE COATING — THIS STATION IS INTENTIONALLY EMPTY ON AN INTEGRAL-COLOR LINE.”

PURPOSE & THE “WHY”

On a decorative Type II line, this is where dye goes into the open pores. **On an integral-color line there is no dye step** — the color was created in the color-anodize tank (Station 07). We keep this position in the manual so you understand the difference and never reach for a dye tank by habit. Depending on your shop’s layout, this tank may simply be **bypassed**, or **used as a hold/rinse position** on the way to seal.



REMEMBER — THERE IS NO DYE ON THIS LINE

Integral color’s whole identity is **color without dye**. If a part is the wrong color, the answer is **never** “dye it” — it’s **strip and re-anodize** to the correct amp-hours on the correct alloy. Adding dye on top of an integral color would only muddy it and would *not* be lightfast.



TECHNICAL STUFF — COULD YOU EVER DYE OVER INTEGRAL COLOR?

Technically the oxide is still porous, so a dye *could* be absorbed — but doing so defeats the purpose: you’d be adding a **fade-prone organic dye** on top of a finish chosen specifically for **lightfastness**. Quality architectural practice does **not** dye integral color. If a brighter or different color is needed, the right answer is a **different route** (dyed Type II for interiors, or two-step electrolytic for durable architectural color).



TIP

When a customer asks for an integral-color part in a color outside the bronze-to-black palette (a bright red, blue, green), the honest answer is: **integral color can’t do that**. Steer them to **dyed Type II** (if interior/decorative) or **two-step electrolytic** (if exterior/durable). Knowing the limits of this process is part of running it well.

SIGN-OFF — STATION 09

Teach-Back: Why there is no dye step on this line; where the color actually came from (Station 07); what the right fix is for a wrong color (strip & re-anodize, not dye); the palette limits of integral color.

“I understand integral color carries no dye step — the color is grown in the anodize tank — and I would never dye an integral-color part to ‘fix’ color.”

Trainee: _____ Trainer: _____ Date: _____

STATION 10 OF 12

SEAL (CLOSE THE PORES)

“THE FINAL LOCK – CLOSE THE PORES FOR DURABILITY AND CORROSION RESISTANCE. THE COLOR IS ALREADY IN.”

PURPOSE & THE “WHY”

Close the open pores of the colored oxide so the coating reaches its full **corrosion and abrasion resistance**. An unsealed anodize is porous — it can absorb stains and corrode through the pores. **Sealing** swells/plugs the pore mouths shut. The classic method is **hot DI water near boiling**, which converts the pore-wall oxide to a swollen hydrated oxide (**boehmite**) that closes the pores. There are also **mid-temperature nickel-acetate** seals and **cold/room-temperature nickel-fluoride** seals — each trading energy, speed, and performance.



REMEMBER — SEAL LOCKS DURABILITY, NOT COLOR

Unlike a dyed part (where seal locks in the *dye*), on an integral-color part the **color is already permanently in the oxide**. Sealing here is about **corrosion resistance, abrasion resistance, and stain resistance** — the architectural durability the finish is famous for. A poor seal lets a beautiful color **corrode and weather** prematurely. An over-sealed/contaminated part can show “**sealing bloom**” (a chalky film) that **dulls the color** — so seal quality still affects appearance.

THE THREE COMMON SEALS

SEAL TYPE	TEMPERATURE	TRADEOFFS
Hot DI water seal	~96–100 °C	Simple, no metals, excellent quality; high energy, slow (~2–3 min/μm), bloom-prone if water is off
Nickel-acetate (mid-temp)	~80–90 °C	Lower energy; excellent corrosion & color retention; widely used in architecture; contains nickel
Cold nickel-fluoride	~25–30 °C	Lowest energy/fastest; often a warm “aging” rinse after; contains nickel + fluoride ; needs tight control



TECHNICAL STUFF — WHY HOT WATER SEALS

Near-boiling water hydrates the pore-wall oxide to **boehmite (AlOOH)**, which occupies more volume and **swells the pore mouths shut**. Nickel seals add nickel salts that precipitate in the pore mouths to help plug them, which is why they work at lower temperature. All three achieve **closed pores** by different routes.



TECHNICAL STUFF — SEALING BLOOM ON A COLORED PART

A powdery white/gray film after sealing (“bloom”) usually comes from **hard water, drag-in, wrong pH, or over-concentrated seal**. On an integral-color bronze or black, bloom is **especially visible** — it dulls and grays the color. The fix is **clean DI water and correct seal chemistry/pH/temperature**, plus a light acid post-dip if needed.



HEADS-UP — NEAR-BOILING TANK = SCALD HAZARD

A hot-water or hot nickel seal runs **~80–100 °C**. Splashes and steam cause **serious scald burns**. Face shield, heat-resistant gloves, long sleeves; mind the steam cloud. Lower parts gently.



HEADS-UP — NICKEL SEALS ARE A HEALTH & WASTE CONCERN

Nickel is a skin sensitizer and the **fluoride** in cold seals is especially hazardous (deep, delayed burns; systemic toxicity). Gloves and goggles, read the SDS, know the fluoride first-aid (**calcium gluconate**), and route nickel/fluoride seal waste to proper treatment — **never to drain**.

MAKE-UP & KEY PARAMETERS

PARAMETER	RANGE / PRACTICE	NOTES
Water	DI (hot-water seal)	Hard water → bloom (dulls color)
Temperature	Hot ~96–100 • Ni-acetate ~80–90 • Ni-F ~25–30 °C	Per chosen seal
pH	Per seal product (often ~5.5–6.5)	Wrong pH = poor seal/bloom
Time	~2–3 min per µm (hot water)	Per seal type & thickness (Class I is thick!)
Result	Closed pores; passes seal-quality test	Dye-stain (B136) / admittance (B457)

STEP-BY-STEP

1. Confirm seal type, temperature, pH, and DI water. 2. Confirm the part came from a clean cool rinse. 3. Immerse gently (mind the scald), full time **for the architectural thickness** (Class I is thick — seal takes longer). 4. Withdraw, rinse if specified, inspect for **bloom** (which dulls the color). 5. Move to final rinse/dry.

TEACH-BACK — CAN YOU ANSWER?

- What sealing does (closes pores) and why (corrosion/abrasion/stain — **color already locked**).
- The three seal types and one tradeoff each.
- The scald hazard; the nickel/fluoride hazard.
- What bloom is and why it dulls an integral color.

TOP MISTAKES

- Under-sealing a thick Class I film (corrodes/weathers).
- Wrong temp/pH; hard water → bloom.
- Plunging into a near-boiling tank.
- Routing nickel/fluoride waste to drain.
- Skipping the seal test.

SIGN-OFF — STATION 10

“I sealed at the correct type/temperature/pH in DI water for the full Class I thickness, checked for bloom, and handled the scald and (if applicable) nickel/fluoride hazards correctly.”

Trainee: _____ Trainer: _____ Seal type/temp: _____

STATION 11 OF 12

RINSE / DRY

“FINISH CLEAN, DRY GENTLY, DON’T UNDO YOUR WORK.”

PURPOSE & THE “WHY”

Give the sealed part a final clean rinse and a **gentle dry** without water spotting or thermal damage. A clean DI final rinse removes seal-tank drag-out so the part dries spot-free — **water spots are glaringly obvious on a dark bronze or black architectural finish** and are pure rework.

PARAMETER	RANGE	NOTES
Final rinse water	DI	Spot-free drying
Rinse temperature	Warm	Aids drying
Dry method	Air / warm forced air; gentle oven	Avoid excessive heat
Handling	By rack / clean gloves	No bare-hand prints on finished architectural parts

**HEADS-UP**
Parts coming off a hot seal are **hot** — burn risk. Let them cool or use gloves. Mind hot racks and dry ovens.

**TIP**
A **DI final rinse** is the cheapest insurance against water spots on a finished bronze/black part. On long architectural extrusions, handle and stack carefully — scuffs and prints on a dark finish show badly.

STEP-BY-STEP

- Final DI rinse.
- Drain.
- Gentle dry.
- Handle by rack/gloves to inspection.

TOP MISTAKES

- Tap-water final rinse (spots).
- Over-aggressive drying.
- Bare-hand handling of finished dark parts.

SIGN-OFF — STATION 11

Teach-Back: Why a clean final rinse + gentle dry protects a dark architectural finish.

“I gave a clean DI final rinse, dried gently without spotting/overheating, and handled finished parts cleanly.”

Trainee: _____ Trainer: _____ Date: _____

STATION 12 OF 12

FINAL INSPECTION

“PROVE IT’S RIGHT – COLOR/ ΔE , THICKNESS, COATING WEIGHT, AND SEAL, TO AAMA 611.”

PURPOSE & THE “WHY”

Verify the finished part meets spec: **color match (ΔE)**, **coating thickness**, **coating weight**, **seal quality**, and **appearance** — typically to **AAMA 611** for architectural anodize. On an integral-color line, **color is the headline check** because there’s no dye to adjust: the part either matches the approved standard or it doesn’t.

CHECK	METHOD	PASS CRITERIA
Color match	Spectrophotometer (ΔE) and/or visual vs. approved standard	Within the spec’d ΔE / range vs. the approved color standard
Coating thickness	Eddy-current gauge (ASTM B244); micro-section if needed	Within spec — architectural Class I $\geq 18 \mu\text{m}$ (0.7 mil)
Coating weight	Mass per unit area (ASTM B137)	Per spec (architectural coatings often spec’d by weight)
Seal quality	Dye-stain (ASTM B136) and/or admittance (ASTM B457)	Low dye pickup / low admittance = well sealed
Appearance	Visual under controlled light	No burns, bloom, pits, streaks, color banding, bad contact marks
Dimensions	Gauging where critical	Within tolerance after dimensional growth



REMEMBER — THE CHECKS THAT MATTER MOST HERE

Color (ΔE) = “does it match the standard?”, eddy-current/coating-weight = “did we grow enough oxide?”, and the seal test = “did we close the pores?” A part can be the right thickness and *still* be the wrong color (wrong alloy/amp-hours) — which is exactly why **color measurement against a standard is non-negotiable on this line**.



TIP — ΔE IN PLAIN TERMS

ΔE is a single number for “how different are these two colors.” A small ΔE means a good match; a large ΔE means visibly off. Architectural specs set a maximum allowable ΔE (and often require parts to fall within a **light/dark color range**, with retained range samples). Measure on the **significant surface** under controlled light.



TIP — THE DYE-STAIN SEAL TEST

Drop a spot of test dye on a sealed area; a **good seal barely picks up color** (wipes clean), a **poor seal stains** (pores still open). The quick visual cousin of the **admittance** test.

SIGN-OFF — STATION 12

Teach-Back: Which check answers “matches standard?” vs. “enough oxide?” vs. “pores closed?”; what ΔE means; what AAMA 611 Class I requires ($\geq 18 \mu\text{m}$).

“I measured color (ΔE) against the approved standard, verified thickness and coating weight, ran the seal test, inspected for defects, and confirmed critical dimensions accounting for growth.”

Trainee: _____ Trainer: _____ ΔE : _____ Thickness: _____ μm

COLOR CONTROL BY THICKNESS & AMP-HOURS

THE IDEAS THAT DON'T LIVE IN ONE TANK — THEY TIE THE WHOLE LINE TOGETHER

Color is set by coating thickness, and thickness is set by amp-hours. At a fixed current density, the longer you anodize the thicker — and darker — the coating: champagne → light bronze → medium → dark bronze → gray → black.

Because thickness tracks **charge per unit area (amp-hours/ft² or A·min/dm²)**, the disciplined way to hit a color is to **run to an amp-hour target and to a physical color standard**, not just a clock time. A part with poor contact draws less current, accumulates fewer amp-hours, and comes out **too light** — which is why racking and amp-hour control are the same conversation.

LEVER	EFFECT ON COLOR	OPERATOR ACTION
Amp-hours (charge)	More = thicker = darker	Run to the amp-hour target for the standard
Current density	Sets the rate of growth	Set current for the part's surface area
Contact quality	Poor contact → fewer amp-hours → too light	Rack tight; verify before energizing
Bath temperature	Warm → burned/off-color	Hold cool & steady; watch cooling
Alloy / temper / lot	Changes the color at the same amp-hours	Run like-with-like; sample first

★

REMEMBER
Three numbers rule this line: **color = thickness = amp-hours** (run to amp-hours and a color standard); **growth ≈ half-in/half-out** (plan tolerances); **Class I ≥18 μm** (architectural durability). Master those and you understand integral color.

⚙

TECHNICAL STUFF — FARADAY IN A HARD HAT
The oxide you grow is proportional to the **charge passed** (Faraday's law). Two parts run for the same minutes can reach different colors if one drew less current (loose contact, low CD). That's why a stopwatch alone is not enough — **integrate the amps over time (amp-hours) and confirm against the physical color standard.**

☀

FUN FACT
Architectural anodizers keep **retained “range” samples** — a lightest-acceptable and a darkest-acceptable bronze — and every production part has to fall *between* them. The color isn't one number; it's a window, and your amp-hours keep you inside it.

DEEP DIVE

ALLOY / TEMPER EFFECTS & LOT-TO-LOT COLOR MATCHING

THE DEFINING CHALLENGE OF INTEGRAL COLOR — THE METAL MAKES THE COLOR

This is the defining challenge of integral color. Because the **alloy's own constituents are part of the colorant**, color depends on **alloy, temper, heat/lot, and even the extrusion press** — not just the bath and the amp-hours.

PRACTICAL CONTROL

- **Rack like-with-like** — same alloy, temper, and ideally lot in one load.
- **Separate extrusions from sheet** — 6063 extrusion and 5005 sheet color differently; run and tune them separately, both to the same **approved color standard**.
- **Sample first** — anodize a coupon of the *actual* metal/lot to the standard before committing production.
- **Control the metal** — specify alloy **and temper** (and sometimes mill) on color-critical architectural work; record source/lot on the traveler.
- **Keep retained range samples** — most architectural specs allow a light-to-dark **color range**; match production within that range.



REMEMBER

The metal makes the color. Mixed alloys/temperatures/lots in one load = a multi-shade load you can't fix without stripping. Lot control and color standards aren't paperwork — they're the product.



HEADS-UP — THE ALL-TOO-COMMON JOB-SITE MISMATCH

An architectural job often combines **extrusions, sheet panels, and castings** on the same building — three mill products that color differently at identical settings. If they aren't each tuned to the **same approved standard**, the finished facade shows visible shade banding. Confirm the standard and the metal source *before* production, not after install.



TECHNICAL STUFF — WHY TWO-STEP IS LESS ALLOY-FUSSY

In two-step electrolytic color the colorant is **metal you deposit yourself** in a second tank, so the base alloy matters far less. That's the core reason the industry drifted toward two-step for new architectural work — you control the color directly instead of relying on the metal to make it. Integral color persists where its proven track record or a specific color is required.



FUN FACT

Some classic buildings were anodized with metal from a **single reserved heat** of aluminum, set aside specifically so every extrusion on the facade would color identically. The finish was demanding enough to dictate how the metal was purchased.

REFERENCE & HONEST ACCOUNTING

THE THREE COLORING ROUTES COMPARED • ENERGY COST

WHEN TO CHOOSE INTEGRAL COLOR — AND WHEN TO POINT THE CUSTOMER ELSEWHERE

	DYE (TYPE II)	INTEGRAL COLOR	TWO-STEP ELECTROLYTIC
Where the color lives	Organic dye in the pores	In the oxide itself	Metal deep in the pores
Lightfastness	Low–medium	Excellent	Excellent
Palette	Any color	Champagne–bronze–gray–black	Champagne–bronze–black
Energy / cost	Low	High	Moderate
Color consistency	Good	Hard (alloy-driven)	Very good
Typical use	Indoor / decorative	Exterior architecture (legacy/spec)	Exterior architecture (modern std)

ENERGY COST — THE HONEST ACCOUNTING

Integral color is **the most energy-intensive of the three coloring routes**:

- **Higher voltage** (~25–75 V vs. ~12–18 V for Type II) → more electrical energy per amp.
- **Longer cycles** to build a thick Class I colored film → more amp-hours.
- **Bigger cooling load** to remove all that energy and hold color-critical temperature.

By contrast, **dyeing** adds only a low-temperature dye dip, and **two-step electrolytic** adds a modest second electrolytic step — both at **lower total energy** than integral color, and two-step matches integral's durability with **better consistency**. This energy/consistency gap is the main reason **two-step electrolytic color has largely displaced integral color** for new architectural work — though integral color is still specified for its proven decades-long track record and certain colors.



REMEMBER

Integral color buys **durability and lightfastness** with **electricity, cooling, and tight alloy control**. When cost or consistency dominates and the color is a bronze/black, **two-step electrolytic** is usually the better tool — know when to point the customer there.



FUN FACT

All three routes can make a bronze that looks identical on day one. Ten years in the sun sorts them out: the **dyed** one fades, while the **integral** and **two-step** bronzes hold — durable ceramic/metal color, not an organic dye.

DEEP DIVE

RACKING, SEAL, RINSING & WATER/DI

THE CROSS-CUTTING PRACTICES THAT PROTECT COLOR, DURABILITY, AND FINISH

• RACKING

The part is the anode, so **the rack is the wire that grows your colored coating**. Firm current-carrying contact in a hidden spot, like-alloy grouping, orientation for gas/drainage, no parts touching, maintained (titanium) racks. Poor contact here means **light/off color** as well as burns and dropped parts. (*Cross-reference Station 01.*)

• SEAL

Closes the pores for **corrosion/abrasion/stain resistance** (color is already locked). Three routes: **hot DI (~96–100 °C)** — metal-free, high energy, bloom-prone; **nickel-acetate (~80–90 °C)** — efficient, excellent retention, contains nickel; **cold nickel-fluoride (~25–30 °C)** — fast/low-energy, nickel + fluoride hazard. Enemies: hard water, wrong pH, contamination → **bloom that dulls the color**. Confirm with **dye-stain (B136)** or **admittance (B457)**.

• RINSING & WATER

Rinses are firewalls between the caustic etch and the acid stages, and protectors of the seal/finish. **Counter-current cascade** saves water; **conductivity** gauges cleanliness. Use **DI** for post-anodize, seal, and final rinses — tap-water **chloride and hardness** cause pitting, uneven seal, and water spots that are obvious on dark finishes.



HEADS-UP

Acidic and caustic streams must be **neutralized under control** (never casually mixed). A **chromate-bearing** or **fluoride/nickel** seal needs **dedicated treatment** — never to drain. Know your shop's segregation scheme.



REMEMBER

Three cross-cutting habits protect every integral-color job: **tight, like-with-like racking** (color & contact), **a clean, correct seal** (durability without bloom), and **DI rinsing** (no pits, no spots). They sound minor — they decide whether a beautiful color survives.



TIP

Treat **DI water** as a color-quality tool, not a utility. The same chloride that pits the anodize and the same hardness that blooms the seal are invisible in tap water — DI removes the variable before it ruins a load.

THE ORGANIC-ACID LOAD

WASTE & ENVIRONMENTAL

SEVERAL REGULATED STREAMS — NEVER TO A FLOOR DRAIN UNTREATED

An integral-color line generates several regulated streams — **never to a floor drain untreated**:

- **Spent sulfuric + organic-acid bath & acid rinses** — neutralize/treat per permit; manage **dissolved aluminum** (precipitated as hydroxide sludge). **The organic acids add organic load (COD/BOD)** to the wastewater — your treatment system and permit must account for it. (Generally lower-hazard than chromic-acid color chemistry, which had hexavalent chromium — but the organic load is a real wastewater consideration.)
- **Spent caustic etch** — high pH; contains dissolved aluminum; neutralize/treat.
- **Desmut waste** — acidic; **fluoride-bearing** desmuts need fluoride treatment (precipitate as calcium fluoride).
- **Nickel-seal waste** (nickel-acetate / nickel-fluoride) — **nickel is regulated**; fluoride needs treatment; segregate and treat — never to sewer.
- **Air** — acid mist (sulfuric/organic) and steam need exhaust/scrubbing per permit.

**HEADS-UP**

Acidic and caustic streams must be **neutralized under control** (never casually mixed); nickel/fluoride streams need **dedicated treatment**; and the **organic-acid load** means your wastewater chemistry is different from a plain Type II line. Know your shop's segregation scheme — the wrong drum or drain can be a reportable violation.

**REMEMBER**

Integral color's environmental footprint is dominated by **energy and the organic-acid load**, plus the usual acid/caustic neutralization and any **nickel/fluoride** seal treatment. It avoids hexavalent chromium, but it is **not** a low-energy, low-load process — segregate and treat every stream by your permit.

**TECHNICAL STUFF — WHY THE ORGANIC ACIDS CHANGE THE WASTEWATER**

The sulfophthalic/sulfosalicylic/maleic/oxalic additives are **organic molecules**, so the spent bath and acid rinses carry a measurable **COD/BOD load** a plain sulfuric line doesn't. Biological or chemical treatment and your discharge permit have to account for that organic content — it's the one wastewater wrinkle unique to integral color.

**FUN FACT**

The historic integral-color processes were prized partly because they color aluminum **without** the hexavalent-chromium color chemistry of their era — a genuine hazard reduction. The trade is that they moved the burden from *chemical toxicity* toward *energy and organic load*.

HOW WE PROVE IT

QUALITY CHECKS & TEST METHODS

THE MAKE-OR-BREAK CHECKS — COLOR/ΔE, THICKNESS/COATING WEIGHT, AND SEAL

CHECK	METHOD (TYPICAL STD)	ANSWERS
Color match	Spectrophotometer ΔE / visual vs. standard	"Right shade & within range?"
Coating thickness	Eddy-current (ASTM B244); micro-section	"Enough oxide grown?"
Coating weight	Mass/area (ASTM B137)	"Enough coating per architectural spec?"
Seal quality	Dye-stain (B136); admittance (B457); acid-dissolution weight loss	"Pores closed?"
Abrasion / wear	Abrasion test per spec	"Durable enough?"
Corrosion	Salt spray (ASTM B117) per spec	"Will it survive service?"
Lightfastness / weathering	Per AAMA 611 / project spec	"Color holds outdoors?"
Appearance/defects	Visual under controlled light	Burns, bloom, pits, streaks, color banding



REMEMBER

Architectural integral color is typically certified to **AAMA 611** (architectural anodize) and **ASTM B580 (architectural Type A)** with **Class I = $\geq 18 \mu\text{m}$ (0.7 mil)** for exterior durability. The make-or-break field checks are **color/ΔE, thickness/coating weight, and seal**.



TIP — MEASURE COLOR LIKE A PRO

Read ΔE on the **significant surface** under **controlled light**, against the **approved standard** — not "by eye" under shop lighting and not part-to-part. Architectural specs set a maximum ΔE and a light/dark range; keep **retained range samples** to judge against.



TECHNICAL STUFF — THICKNESS, WEIGHT, AND SEAL TOGETHER

No single number proves the part. **Thickness/coating weight** says you grew enough oxide; the **seal test** says the pores are closed; and **color/ΔE** says the shade is right. A part can pass thickness and still fail color (wrong alloy/amp-hours) — which is why color is non-negotiable on this line.

FIELD REFERENCE

TROUBLESHOOTING
QUICK-INDEX

FIND YOUR SYMPTOM, READ THE LIKELY CAUSE, TRY THE FIX

DEFECT / SYMPTOM	LIKELY CAUSE	FIX
Color too light	Too few amp-hours; poor contact; low CD; warm bath	Run to amp-hours; re-rack tight; check CD/cooling
Color too dark	Too many amp-hours; bath off-spec	Reduce amp-hours; verify bath
Color mismatch part-to-part	Mixed alloys/tempers/lots ; inconsistent dwell/amp-hours/temp	Rack like-with-like; standardize; sample first
Extrusion vs. sheet won't match	Different mill products color differently	Separate tuned runs; match to standard
Color banding / streaks	Uneven etch/desmut; uneven agitation; smut left on	Verify pretreat uniformity; check agitation
Burning (powdery, dark, rough)	Bath too warm; CD too high; poor agitation/cooling	Restore cooling/agitation; lower CD
Thin/no coat (and too light)	Poor contact; wrong polarity ; low current; low acid	Re-rack; verify part = anode ; check current/acid
Pitting	Chloride ; impurities	Lab-check chloride; DI; treat bath
Dull / muddy color	High dissolved Al; off-spec organic acid; high-Cu/Si metal	Lab-check bath; confirm alloy
Poor seal (corrodes, stains, fails test)	Wrong seal temp/pH; contaminated seal; under-sealed thick film	Correct temp/pH/DI; re-seal full time
Sealing bloom (chalky, dulls color)	Hard water; wrong pH; over-concentrated seal	DI water; correct pH/conc.; light post-dip
Water spots on dark finish	Tap-water final rinse	DI final rinse



TIP

Most integral-color defects trace to one of four roots: **amp-hours/contact** (too light/dark, banding), **temperature/cooling** (burning), **alloy/lot** (mismatch, muddy), or **contamination/seal** (pitting, bloom, spots). Name the root, not just the symptom.



REMEMBER

Two of these are **safety-relevant** as well as quality: **wrong polarity** (verify the part is the anode) and anything that increases acid attack or heat. When in doubt about a bath change, ask your supervisor first.

— EVERY TERM, PLAIN AND SIMPLE

GLOSSARY

PLAIN-LANGUAGE DEFINITIONS FOR EVERYTHING IN THIS MANUAL

Admittance test: electrical test (ASTM B457) of how “open” the coating still is — confirms sealing (lower = better seal).

AAMA 611: the architectural-aluminum anodize specification (color, thickness/class, performance) used for building products.

Amp-hours (A·h): the charge passed = current × time. **On this line, amp-hours set thickness and therefore color.**

Anode: the **positive** electrode. In anodizing it’s **your part** — where the colored oxide grows.

Anodizing: growing an integral oxide layer out of a metal’s surface by making the part the anode in an electrolyte.

Barrier layer: the thin, dense oxide at the bottom of the coating, against the metal.

Boehmite: hydrated aluminum oxide (AlOOH) formed during hot-water sealing; its swelling closes the pores.

Cathode: the **negative** electrode (lead or aluminum) — carries current, bubbles off hydrogen; does **not** feed the coating or color.

Class I / Class II: architectural coating thickness classes — **Class I ≥18 μm (0.7 mil)** for exterior; Class II ~10–18 μm for less demanding exposure.

Coating weight: mass of oxide per unit area (ASTM B137) — an architectural acceptance measure.

Current density (ASF / A/dm²): amps per unit area.

Desmut / deox: an acid dip that removes the dark **smut** left by caustic etching.

Dimensional growth: the coating grows **~half in, half out**; each surface moves out ~half the thickness.

Duranodic / Kalcator: historic trade names for integral-color processes (sulfosalicylic / sulfophthalic acid + sulfuric).

ΔE (Delta-E): a single-number measure of color difference vs. a standard; small = good match.

Drag-out / drag-in: solution clinging to a part leaving a tank (out) / contaminating the next (in).

Eddy-current gauge: non-destructive instrument for measuring coating thickness on aluminum (ASTM B244).

Etch: a hot **caustic** dip that removes natural oxide and a little aluminum, giving the **architectural satin** finish.

Integral color: color produced **by the anodizing itself** (no dye), from occluded alloy constituents + the organic-acid-modified oxide; **lightfast** bronzes/gray/black.

Lightfastness: resistance of a color to fading under UV/sunlight — integral color’s signature strength.

Organic acids (sulfophthalic / sulfosalicylic / maleic / oxalic): additives to the sulfuric bath that harden and color the oxide.

Porous layer: the thick upper oxide full of vertical pores (here the color is in the oxide walls, not added to the pores).

Sealing: the final step that **closes the pores** (hot water / nickel-acetate / cold nickel-fluoride) for corrosion/abrasion/stain resistance.

Sealing bloom: a chalky film after sealing (hard water / wrong pH / contamination) that **dulls the color**.

Significant surface: the surface that must meet color/thickness/finish spec; where you measure.

Smut: dark residue of alloying elements (Cu, Si, Fe, Mn) left by caustic etching.

Two-step electrolytic color: anodize then deposit metal salts (tin/nickel/cobalt) in the pores — durable, consistent, the modern architectural standard.

Valve metals: metals that anodize to protective oxides (aluminum, titanium, magnesium, tantalum, niobium).

Water-break test: a free cleanliness check — clean metal sheets water; dirty metal beads it.



REMEMBER

Five terms matter most: **anode** (your part), **integral color** (color from the coating itself), **amp-hours = thickness = color, alloy/temper/lot** (the color driver), and **AAMA 611 / Class I** (the architectural target). Master those and you understand integral color.



FUN FACT

“Integral” means *part of the whole* — and that’s the whole story of this finish: the color isn’t on the part, it *is* the part’s surface. That’s why it lasts the life of a building.

— TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

HOW A SUPERVISOR PROVES — ON PAPER — THAT EACH OPERATOR IS TRAINED AND SIGNED OFF BEFORE RUNNING SOLO

Sign-off levels: **O** = Observed only · **A** = Assisted under supervision · **Q** = Qualified to run solo · **T** = Qualified to Train others.

#	STATION / SKILL	O/A/Q/T	TRAINER INIT & DATE	OPERATOR INIT & DATE	RE-CERT DUE
1	Safety / PPE / SDS / hygiene — acid + organic acid, caustic, hot seal, hydrogen, higher-V DC (all stations)	—	—	—	—
2	Station 01 — Inspection & Racking (contact + alloy/temper/lot)	—	—	—	—
3	Station 02 — Clean / Degrease + water-break	—	—	—	—
4	Station 03 — Etch (caustic, architectural satin)	—	—	—	—
5	Station 04 — Rinse + conductivity	—	—	—	—
6	Station 05 — Desmut / Deox	—	—	—	—
7	Station 06 — Rinse	—	—	—	—
8	Station 07 — Color-Anodize (part = anode) operation	—	—	—	—
9	Color control by amp-hours/thickness + alloy/lot matching	—	—	—	—
10	Dimensional growth & masking	—	—	—	—
11	Station 08 — Post-anodize rinse	—	—	—	—
12	Station 09 — (No dye — understand why)	—	—	—	—
13	Station 10 — Seal (close pores)	—	—	—	—
14	Station 11 — Rinse / Dry	—	—	—	—
15	Station 12 — Final Inspection (color/ΔE, thickness, coating weight, seal, AAMA 611)	—	—	—	—
16	LOTO on rectifier / higher-voltage electrical safety	—	—	—	—
17	Emergency response — eyewash, shower, spill, fluoride first-aid	—	—	—	—
18	Waste segregation & environmental (incl. organic-acid load)	—	—	—	—
19	Troubleshooting — symptom recognition	—	—	—	—

Operator name: _____ Hire/start date: _____

Supervisor name: _____ Review cycle: ☐ Annual ☐ Other: _____



HEADS-UP

Safety rows (1, 16, 17, 18) are not optional and not “fill in later.” An operator may not work *any* station until PPE/SDS/hygiene, LOTO (higher-voltage), emergency response (including **fluoride first-aid** where used), and waste-control awareness are signed. Safety competency precedes production competency.

PART FOUR — CERTIFICATION

20 QUESTIONS • PASSING SCORE 80% (16/20)

COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED



REMEMBER

Passing score: 80% (16 of 20). Any **safety** question answered wrong (**Q3, Q9, Q13, Q17, Q19**) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: _____ Date: _____ Score: _____ / 20 **Safety-gate Qs: 3, 9, 13, 17, 19**

Q1. On an integral-color line, the color is produced by:

- A. Soaking organic dye into the pores after anodizing B. Painting the part after sealing C. The anodizing itself — the color is grown into the oxide in one step, with no dye D. Adding pigment to the seal tank

Q2. Anodizing differs from electroplating mainly because in anodizing the part is the:

- A. Anode, and an oxide is grown out of the part itself B. Cathode, and a different metal is deposited on it C. Insulator, and nothing happens D. Cathode, and it dissolves

Q3. [SAFETY GATE] Compared with a Type II line, an integral-color line raises two hazards in particular. They are:

- A. Radiation and noise B. Only the lighting C. Nothing changes D. A higher-voltage DC supply (more shock/arc energy) and a larger heat/cooling load

Q4. On this line, the color is made **darker** by:

- A. Using a stronger dye B. Growing a thicker coating — i.e., running more amp-hours C. Sealing longer D. Lowering the voltage

Q5. Which describes the integral-color electrolyte?

- A. Pure water B. Sodium hydroxide C. Chromic acid only D. Sulfuric acid blended with one or more organic acids (e.g., sulfophthalic/sulfosalicylic)

Q6. (Short answer) Explain in your own words why anodizing is **not** plating — name which electrode the part is and **where the coating comes from**.

Q7. The integral-color palette is essentially limited to:

- A. Bright reds, blues, and greens B. White only C. Champagne, bronzes, gray, and black D. Any color you want

Q8. Why is integral color the preferred anodize for **building facades and curtain walls**?

- A. It's the cheapest finish B. Its color is in the ceramic, so it's extremely lightfast and survives decades of UV/weather C. It's the only anodize that exists D. It needs no electricity

Q9. (Short answer) [SAFETY GATE] Name the **two corrosive hazards** on the line (one high-pH, one low-pH), and state the universal rule for adding acid during bath make-up.

Q10. Of the three coloring routes, which is generally the **most energy-intensive**?

- A. Integral color (high voltage, long cycle, big cooling load) B. Dyeing (Type II + dye) C. Two-step electrolytic color D. They're all identical

Q11. A loaded rack of mixed alloys/temperatures comes out of the integral-color tank in several different shades. The most likely cause is:

- A. The seal was too hot B. The alloy/temper/lot makes the color, so mixed metals color differently at the same settings C. Too much dye D. The rectifier was off

Q12. (Short answer) What is **smut**, where does it come from, and which station removes it? Why does consistent desmut matter for **color** on this line?

Q13. [SAFETY GATE] A hot-water or nickel-acetate seal tank runs at about **80–100 °C**. The main hazard is:

A. Electrical shock B. Radiation C. Scald / steam burns D. None

Q14. A finished integral-color part is the **wrong color**. The correct remedy is:

A. Strip and re-anodize to the right amp-hours on the right alloy B. Dip it in dye to fix it C. Paint over it D. Seal it again

Q15. Architectural integral color is typically specified at **Class I**, which means a coating thickness of about:

A. 0.5–1 µm B. 1–2 mm C. There is no thickness requirement D. ≥18 µm (0.7 mil)

Q16. (Short answer) Name the **three** common seal methods and give **one tradeoff** of each. On an integral-color part, what does sealing lock in — the color, or the durability?

Q17. (Short answer) [SAFETY GATE] List **three** pieces of PPE required at the sulfuric/organic-acid anodize and desmut tanks, and name the special first-aid concern if your desmut or cold seal contains **fluoride**.

Q18. Sealing **bloom** (a chalky film) on a dark bronze part is usually caused by:

A. Too little voltage during anodizing B. The part being too clean C. Hard water, wrong pH, or over-concentrated/contaminated seal D. Using DI water

Q19. (Short answer) [SAFETY GATE] Name the **flammable gas** produced at the cathode during anodizing and **two** controls that keep it from becoming a fire hazard.

Q20. (Short answer) Briefly contrast the **three ways to color anodized aluminum** (dye, integral, two-step) — one defining trait of each.



SAFETY SECTION — MUST BE CORRECT TO CERTIFY

Questions 3, 9, 13, 17, and 19 cover hazards that can permanently harm you or others. A miss on any safety item means re-teach and re-test before sign-off.

End of test. Hand to your trainer for grading.

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD



HEADS-UP

Keep this page out of the trainee's copy. Questions **3, 9, 13, 17, and 19** are safety-critical — any wrong answer there requires re-training on that topic before sign-off, regardless of total score.

- Q1. C** — The color is grown **by the anodizing itself, in one step, with no dye**.
- Q2. A** — The part is the **anode**, and oxide is grown out of the part itself.
- Q3. D** — A **higher-voltage DC supply** (more shock/arc energy) and a **larger heat/cooling load** than Type II.
- Q4. B** — Darker = **thicker coating = more amp-hours**.
- Q5. D** — **Sulfuric acid + organic acid(s)** (sulfophthalic/sulfosalicylic/maleic/oxalic).
- Q6.** In plating the part is the **cathode** and a **different metal is deposited onto it**. In anodizing the part is the **anode**, and the coating is **aluminum oxide grown out of the part's own surface** (oxygen from the bath + the part's aluminum) — nothing is deposited from elsewhere.
- Q7. C** — Champagne, bronzes, gray, and black (no bright colors).
- Q8. B** — The color is in the **ceramic**, so it's **extremely lightfast** and survives decades of UV/weather.
- Q9.** The **caustic etch** (high pH / sodium hydroxide) and the **sulfuric + organic-acid anodize/desmut** (low pH) — both cause severe burns. Rule: **always add ACID to WATER, never water to acid**.
- Q10. A** — **Integral color** (high voltage, long cycle, big cooling load) is the most energy-intensive of the three routes.
- Q11. B** — The **alloy/temper/lot makes the color**, so mixed metals color differently at the same settings. (Rack like-with-like.)
- Q12.** Smut is the dark film of alloying elements (Cu, Si, Fe, Mn) **left by the caustic etch**, removed at the **desmut/deox** station. Consistent desmut matters because those constituents are the **same elements that color the coating** — uneven desmut → uneven color (and there's no dye to mask it).
- Q13. C** — Scald/steam burns (seal runs ~80–100 °C). Face shield, heat gloves, mind the steam.
- Q14. A** — **Strip and re-anodize** to the right amp-hours on the right alloy. (Never dye an integral-color part to “fix” it.)
- Q15. D** — **≥18 μm (0.7 mil)** = architectural Class I.
- Q16.** Hot DI water (~96–100 °C) — simple/metal-free but high energy and bloom-prone; **nickel-acetate (~80–90 °C)** — efficient/excellent retention but contains nickel; **cold nickel-fluoride (~25–30 °C)** — fast/low-energy but nickel + fluoride hazard. On an integral-color part, sealing locks in **durability** (corrosion/abrasion/stain resistance) — the **color is already locked into the oxide**.
- Q17. PPE (any three):** sealed chemical splash goggles, face shield, chemical-resistant gloves, apron/suit, chemical-resistant boots. **Fluoride concern:** fluoride causes **deep, delayed, systemically toxic burns** — requires specific first-aid (e.g., **calcium gluconate** per SDS/medical plan) and immediate medical attention; ordinary flushing alone may not be enough.
- Q18. C** — Hard water, wrong pH, or over-concentrated/contaminated seal. (DI water + correct chemistry prevents it.)
- Q19.** **Hydrogen** gas (at the cathode). Controls (any two): **ventilation/exhaust running, no open flames/sparks/grinding nearby**, no smoking, keep ignition sources away.
- Q20.** **Dye** = organic dye **in the pores** (cheap, any color, fades). **Integral** = color **in the oxide itself** (durable/lightfast, bronzes-to-black, energy-hungry, alloy-fussy). **Two-step** = **metal deposited in the pores** (durable, consistent, the modern architectural standard).

Answer-key letter distribution (MC): A: Q2, Q10, Q14 · B: Q4, Q8, Q11 · C: Q1, Q7, Q13, Q18 · D: Q3, Q5, Q15 — a healthy spread across all four letters.



REMEMBER

A score of 16/20 passes — but every safety question (3, 9, 13, 17, 19) must be correct to certify. Miss a safety item → re-teach and re-test before sign-off. We don't gamble with people.

PRINT, FILL IN, SIGN



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CERTIFICATE OF COMPLETION
INTEGRAL COLOR ANODIZING TRAINING PROGRAM

This certifies that

OPERATOR NAME

has successfully completed the **Integral Color Anodizing** training course — covering the full workflow (Inspection & Racking, Clean, Etch, Rinse, Desmut/Deox, Color-Anodize, Seal, Rinse/Dry, and Final Inspection), the fundamental concept that **the part is the anode and the oxide is grown out of the part itself**, the defining principle that **the color is produced by the anodizing itself (no dye) and deepens with coating thickness / amp-hours**, the critical role of **aluminum alloy and temper in color and lot-to-lot matching**, the three coloring routes (dye, integral, two-step electrolytic), dimensional growth, color control by amp-hours, electrical-contact and racking practice, sealing chemistry and seal-quality testing, color/ ΔE measurement, coating thickness/weight, and **AAMA 611 architectural acceptance** — and, above all, the **safety practices** required of every operator (sulfuric + organic-acid and caustic-etch burn control, higher-voltage DC controls, hot-seal scald control, hydrogen control, fluoride first-aid where applicable, spill response, and waste segregation) — and has passed the Completion Test with a score of _____ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: _____ Re-certification due: _____

TRAINEE SIGNATURE

CERTIFYING TRAINER

SUPERVISOR

Training reference only. Verify all parameters against your chemistry supplier's TDS, customer/architectural specifications (e.g., AAMA 611, ASTM B580/B137/B244), and applicable regulatory requirements before production use. Sulfuric and organic acids and caustic etch cause severe burns; the process runs at higher DC voltage with a large heat load; seal tanks run near boiling — always consult current SDS and comply with OSHA, EPA, REACH, and local regulations.

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